



Module	Portfolio	Assessment Type
Collaborative Development (5CS024)	2	Individual Report

VROOM Car Rental Services – Business Analyst

Student Id 2329860

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Section :L5CG5

Group :L5CG5 (VROOM)

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Lecturer : Mr. Biraj Dulal

Tutor :Mr. Ashish Acharya

Acknowledgment

The process of making the car rental website project has been a very rewarding experience from professional, academic, and technical aspects as they have been aiding in my improvement of the respective aspects. Thus, for this I would like to express my sincere gratitude towards the module itself, the module leader of collaborative development Mr. Biraj Dulal teacher for giving me this opportunity to write the report on the work done during the project. I would also like to express my thanks to Mr. Ashish Acharya who guided me through the whole project and corrected the mistakes made during the project. His guidance during the semester and project has been a crucial aspect of the completion of the project. Likewise, I am grateful to Herald College Kathmandu for including this module in our studies and for all the college resources used.

Similarly, all the direct and indirect support from friends and peers too has been helpful to finish the project. My team members too have helped me immensely; Developers and Project Managers were a vital part of the project thus, I am very thankful for their help and collaboration during the time of the project.

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Self-appraisal form

Student number	2329860	Name	Karan Sah
Project	VROOM – CAR RENTAL SYSTEM	Date	20 th May, 2024
Role	Business Analyst	Team	L5CG5 (Team VROOM)
Sprint	2		

Personal objectives – performance measurement

Objectives	Evidence provided	Evaluation <i>Student / tutor</i>	
Functional Requirement and Design	As done in Sprint 1, the same product backlog was used in Sprint 2 nevertheless the priorities were changed considering the real-time changes in the project. The priorities, acceptance criteria, and user stories were included in the product backlog through research and according to client requirements. The designs were done to facilitate development which covers the system of car rental. Use Case, ERD, activity, sequence, and wireframe of websites were made. The functional requirements and designs' product backlog, UML diagrams, wireframes are all included in Appendix A. Links: Product Backlog/ User Stories Sprint 2 Backlog		

	Designs Wireframes UML ERD		
<i>Tutor feedback:</i>			
Functional Testing	The testing was done by simulating the behaviour of the targeted users i.e. users renting car and the admin managing the page. The features were tested accordingly manually. The most functionalities of the product was tested according to the black box testing principles. The application was tested in chrome as well as local host. Most of the web application ran according to the backlog nevertheless there were few errors and bugs. The evidences include the screenshots of test cases and bug report in Appendix A. Link: Bug Report Test Cases Screenshot of GitHub Issues		
<i>Tutor feedback:</i>			

		48/50	/50
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Evidence of good collaboration

Good communication and file sharing

A good collaboration was required for the project as each member had distinct different roles. Thus, for this purpose, Jira was used as a sprint backlog. The wireframes, UML diagrams, and ERD were shared through Google chats after their completion. They were discussed in online and offline meetings too.

The conflicts occurred, and the design and development processes were all discussed and figured out in online and offline meetings.

Evidence includes links to offline and online meetings, files shared in Google chats

Link:

[Offline Meetings](#)

[Google Chats](#)

[Files](#) [Shared](#) [through](#) [JIRA](#)
[Online Meetings](#)

Continuing Personal Development (CPD)

There were different stages where I was lost during the project nevertheless with continuous personal development through discussion with teammates, teacher, and client and through self-study all the ignorance on the topic was alleviated. The challenges included conflicts between teammates and less knowledge of the topic which was solved.

I grew professionally, personally, academically, and technically. There was a lot of growth in the before-mentioned terms.

Evidences include:

[Report on Continuous Personal Development](#)

[YouTube Channel for Solving Conflict between Teammates in a Project](#)

[YouTube Channel for Business Analysts](#)

[Meeting Minutes](#)

[YouTube Channel for Collaborative Development](#)

Issue tracking

The issues in sprint 2 were as minor as in sprint 1. They had minimalistic effects on the overall project which were solved quickly after they were listed in Jira as issues and in GitHub issues. The issues were discussed during our classes too. Likewise, few meetings were held to address the issues.

Issues were on the sign-up page, adding a driver, changing the password, and minor CSS Issues.

Issue-tracking screenshots are included in Appendix B.

[Report on Issue Tracking](#)

[Screenshot of Issue of User Change Password in GitHub](#)

[Screenshot of Issue of Automatic Sign out in GitHub](#)

[Screenshot of Overall Issues in GitHub](#)

[Screenshot of Sign Up button not working in Jira](#)

Appendix A

Product Backlog/User Stories

Priority	Requirement Number	Epic	User Story	Acceptance Criteria
1	A1	Admin	As an admin, I want to list and view all cars listed on the platform, so that I can manage the inventory efficiently.	- Admin dashboard displays a list of all cars with relevant details.
2	A2	Admin	As an admin, I want to be able to update car information, so that I can keep the car listings accurate and up-to-date.	- Admin can access and modify car details including specifications and images.
3	A3	Admin	As an admin, I want to be able to add new cars to the system, so that I can expand the inventory.	- Admin interface allows for the addition of new cars with all necessary details.

4	A4	Admin	As an admin, I want to be able to remove cars from the system, so that	- Admin can delete car listings from the system.
---	----	-------	--	--



			outdated or unavailable cars can be taken off the platform.	
5	A5	Admin	As an admin, I want to view cars that are currently booked, so that I can manage bookings effectively.	- Admin dashboard displays a list of booked cars with relevant booking details.

6	A6	Admin	As an admin, I want to view booking history of customers, so that I can track usage patterns and analyze performance.	- Admin dashboard provides access to comprehensive booking history records.
7	A7	Admin	As an admin, I want to view all the customers registered on the platform, so that I can manage user accounts and interactions.	- Admin dashboard displays a list of all registered customers.
8	A8	Admin	As an admin, I want to view details of individual customers, so that I can provide personalized assistance and support.	- Admin interface allows access to customer profiles with relevant information.

10	A10	Admin	As an admin, I want to view drivers associated with the platform, so that I can manage driver resources effectively.	- Admin dashboard displays a list of registered drivers.
11	A11	Admin	As an admin, I want to view	- Admin dashboard



			drivers who are currently on a drive, so that I can track driver availability.	provides real-time information on active drivers.
--	--	--	--	---

12	A12	Admin	As an admin, I want to be able to add new drivers to the system, so that driver resources can be expanded as needed.	- Admin interface allows for the addition of new drivers with relevant details.
13	A13	Admin	As an admin, I want to be able to remove drivers from the system, so that inactive or terminated drivers can be managed appropriately.	- Admin can delete driver profiles from the system.

15	U1	User	As a user, I want to view a dashboard upon logging in, so that I can easily access relevant information and features.	- User dashboard provides an overview of available cars, booking status, and other pertinent details.
16	U2	User	As a user, I want to view cars grouped by brand and type, so that I can quickly find options that match my preferences.	- User interface includes filtering options to group cars by brand and type for easy browsing.

17	U3	User	As a user, I want to be able to get loyalties for my continued usage of the platform, so that I can benefit from	- Loyalty program automatically tracks user activity and rewards loyalty points accordingly.
----	----	------	--	--



			loyalty rewards.	
--	--	--	------------------	--

18	U4	User	As a user, I want to be able to sign up for an account, so that I can access booking functionalities and personalized features.	- User registration process allows for the creation of new accounts with required information.
19	U5	User	As a user, I want to be able to log in to my account, so that I can access my profile and booking history.	- User authentication system verifies credentials and grants access to user accounts.
20	U6	User	As a user, I want to be able to book a car for my desired dates, so that I can plan my trips in advance.	- Booking interface allows users to select desired dates and car options, with availability checks.

23	U9	User	As a user, I want to be able to update and edit my profile information, so that my account details are accurate and up-to-date.	- Profile management interface allows users to modify personal information and preferences.
24	U10	User	As a user, I want to be able to view my	- User dashboard provides



			booking history, so that I can track my past trips and expenditures.	access to comprehensive booking history records.
--	--	--	--	--

25	U11	User	As a user, I want to be able to contact the website through the "Contact Us" feature, so that I can get assistance or provide feedback.	- "Contact Us" page includes a form or contact details for user inquiries and feedback.
26	U12	User	As a user, I want to be able to contact customer care for assistance or support, so that I can resolve any issues promptly.	- User interface provides access to customer care contact information or support ticket submission.

Figure 1 Product Backlog/ User Stories

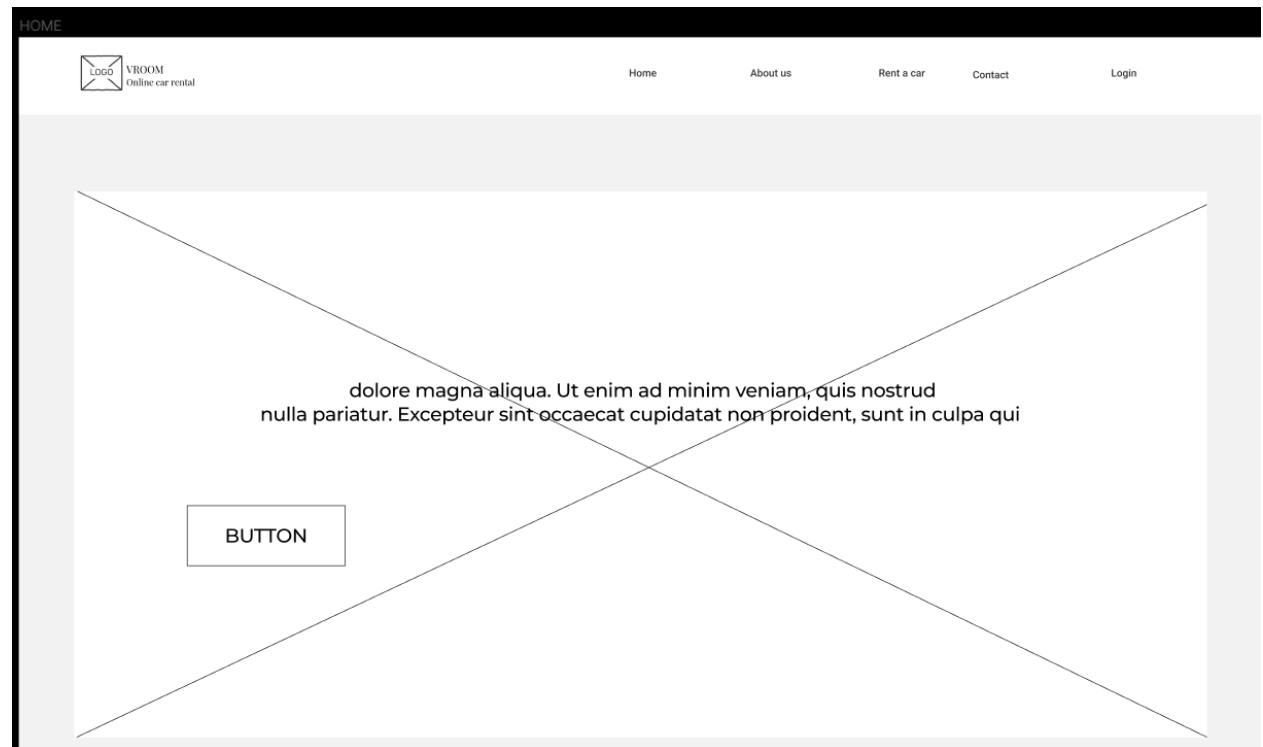
[Back to Table](#)

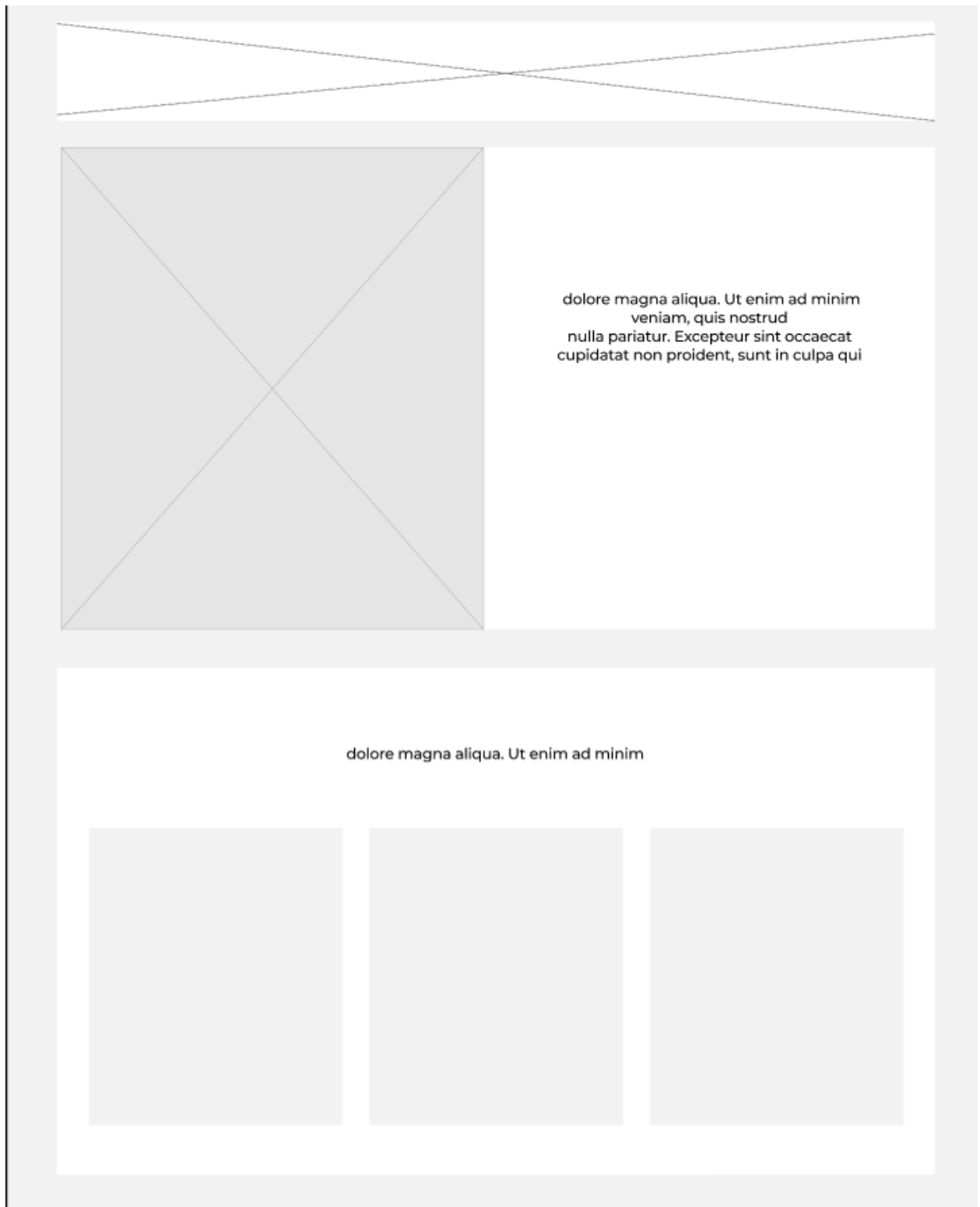
Sprint 2 Backlog

VRM Sprint 2 23 Apr – 7 May (32 issues)			0	0	0	Complete sprint	...
Complete the system							
✓ VRM-110	In the wireframe of the view all cars page of the admin panel add, a edit button so that the admin can ...	DONE ✓	-	KS			
✓ VRM-109	Make the wireframe of the add new car for the admin page	DONE ✓	-	KS			
✓ VRM-108	Make the wireframe of the adminViewCar page	DONE ✓	-	KS			
VRM-88	As an admin, I want to be able to remove cars from the system, so that outdated or unavailable cars ca...	DONE ✓	-				
VRM-85	-As an admin, I want to view all cars listed on the platform, so that I can manage the inventory efficiently.	IN PROGRESS ▾	-				
✓ VRM-107	Make the frontend of Manage Cars in the admin Section with column of available car and a popup whe...	DONE ✓	-	KS			
✓ VRM-94	Make the adminViewCar page by the use of frontend languages	IN PROGRESS ▾	-				
✓ VRM-95	Make the database to store the car list and view them in the inventory	IN PROGRESS ▾	-				
VRM-86	As an admin, I want to be able to update car information, so that I can keep the car listings accurate an...	IN PROGRESS ▾	-				
✓ VRM-98	Make the wireframe for update the information of the car for the admin page	IN PROGRESS ▾	-	PK			
✓ VRM-114	Using the frontend language add a button to edit the car for the admin, for this take the reference of t...	TO DO ▾	-				
✓ VRM-115	In the edit button made by the use of frontend language, add the code so that the admin can edit the ...	TO DO ▾	-				
✓ VRM-111	Write the code in such a way that when the admin updates the info of the car, then it gets updated in ...	TO DO ▾	-				
VRM-87	As an admin, I want to be able to add new cars to the system, so that I can expand the inventory.	TO DO ▾	-				
✓ VRM-96	Taking the reference of the wireframe make the addNewCar page by the use of frontend languages	TO DO ▾	-				
✓ VRM-97	Create a database to store the details of new car when admin adds new car into the system	TO DO ▾	-				
✓ VRM-100	In the wireframe of the view all cars, add a button to remove the car.	TO DO ▾	-	PK			
✓ VRM-101	Add a delete button using frontend language to remove car, take the reference of the wireframe.	TO DO ▾	-				
✓ VRM-112	In the delete button of the frontend, add the code to delete the car from the database so that the spe...	TO DO ▾	-				
✓ VRM-113	Before removing the car from the database, make a pop up button to ask the admin if they really want...	TO DO ▾	-				
VRM-89	As an admin, I want to view all the customers registered on the platform, so that I can manage user acc...	TO DO ▾	-				
VRM-90	As an admin, I want to be able to add new drivers to the system, so that driver resources can be expand...	TO DO ▾	-				
✓ VRM-102	Make the wireframe of the add drivers page and also make the frontend of the page using the langua...	TO DO ▾	-				
✓ VRM-103	Write a SQL query to create a database table named drivers where the admin can add, remove and up...	TO DO ▾	-				
VRM-91	As an admin, I want to be able to give loyalties to customers, so that customers retention and satisfacti...	TO DO ▾	-				
✓ VRM-105	Make a wireframe for the loyalties to be provided for the customers	TO DO ▾	-	KS			
✓ VRM-106	Make the loyalties page using frontend language	TO DO ▾	-	KS			
VRM-92	As a user, I want to be able to log in to my account, so that I can access my profile and booking history.	TO DO ▾	-				
✓ VRM-104	Make wireframe of the booking history page for the users and create the page using frontend languag...	DONE ✓	-				
VRM-93	As a user, I want to be able to book a car for my desired dates, so that I can plan my trips in advance.	TO DO ▾	-				
✓ VRM-99	Write SQL query to update the information of the car	TO DO ▾	-				
VRM-116	Sign Up Button not Working	TO DO ▾	-				
+ Create issue							

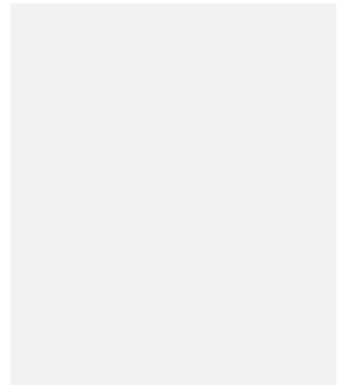
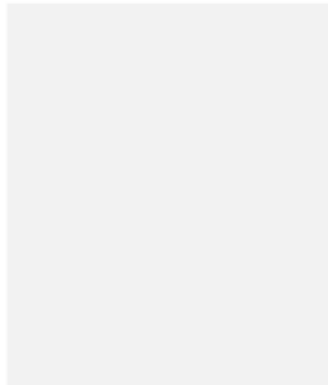
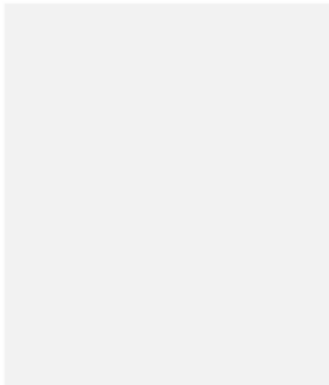
[Back to Table](#)

Wireframes

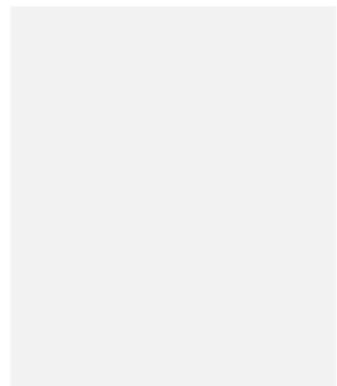
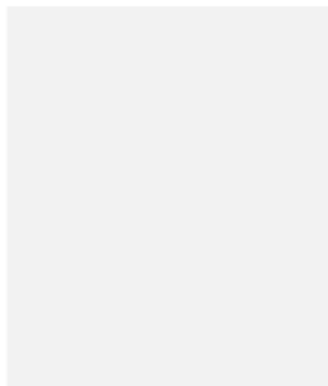
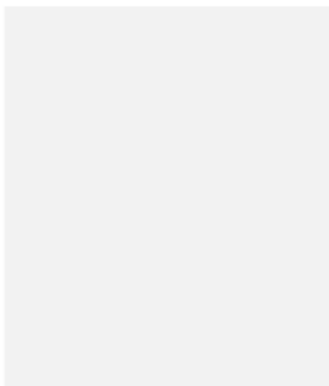




dolore magna aliqua. Ut enim ad minim



dolore magna aliqua. Ut enim ad minim



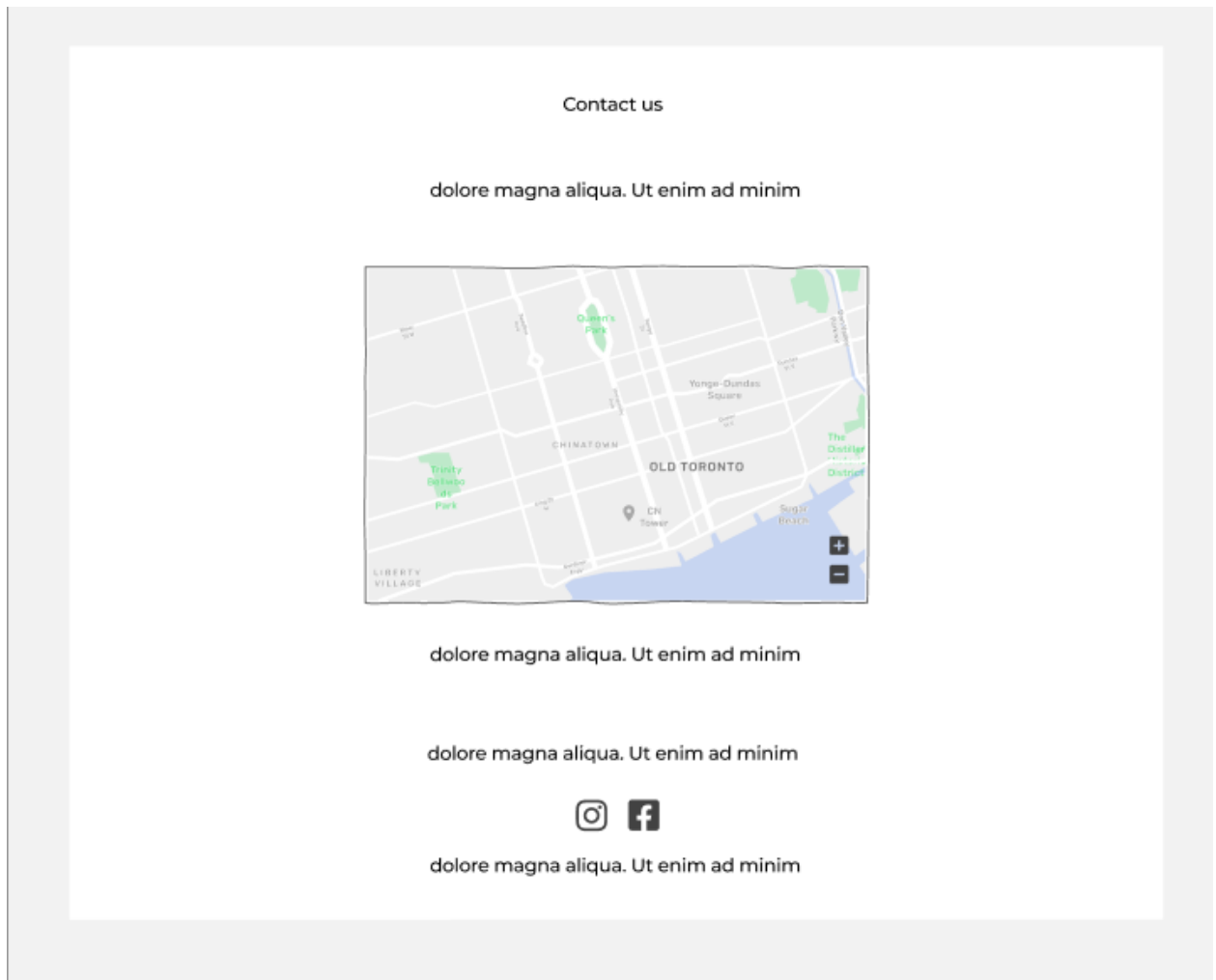


Figure 2 Home Page Wireframe

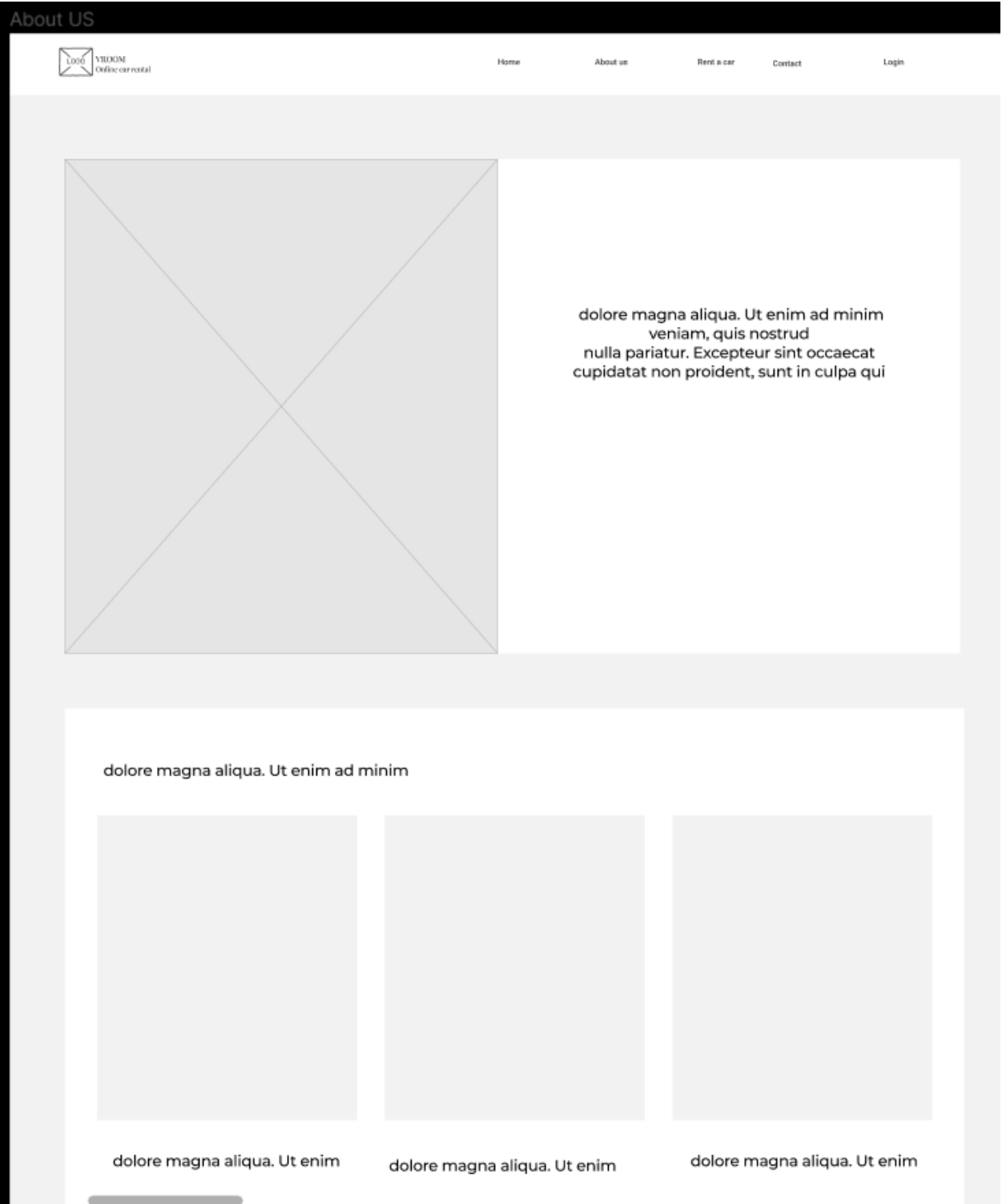


Figure 3 About Us Wireframe

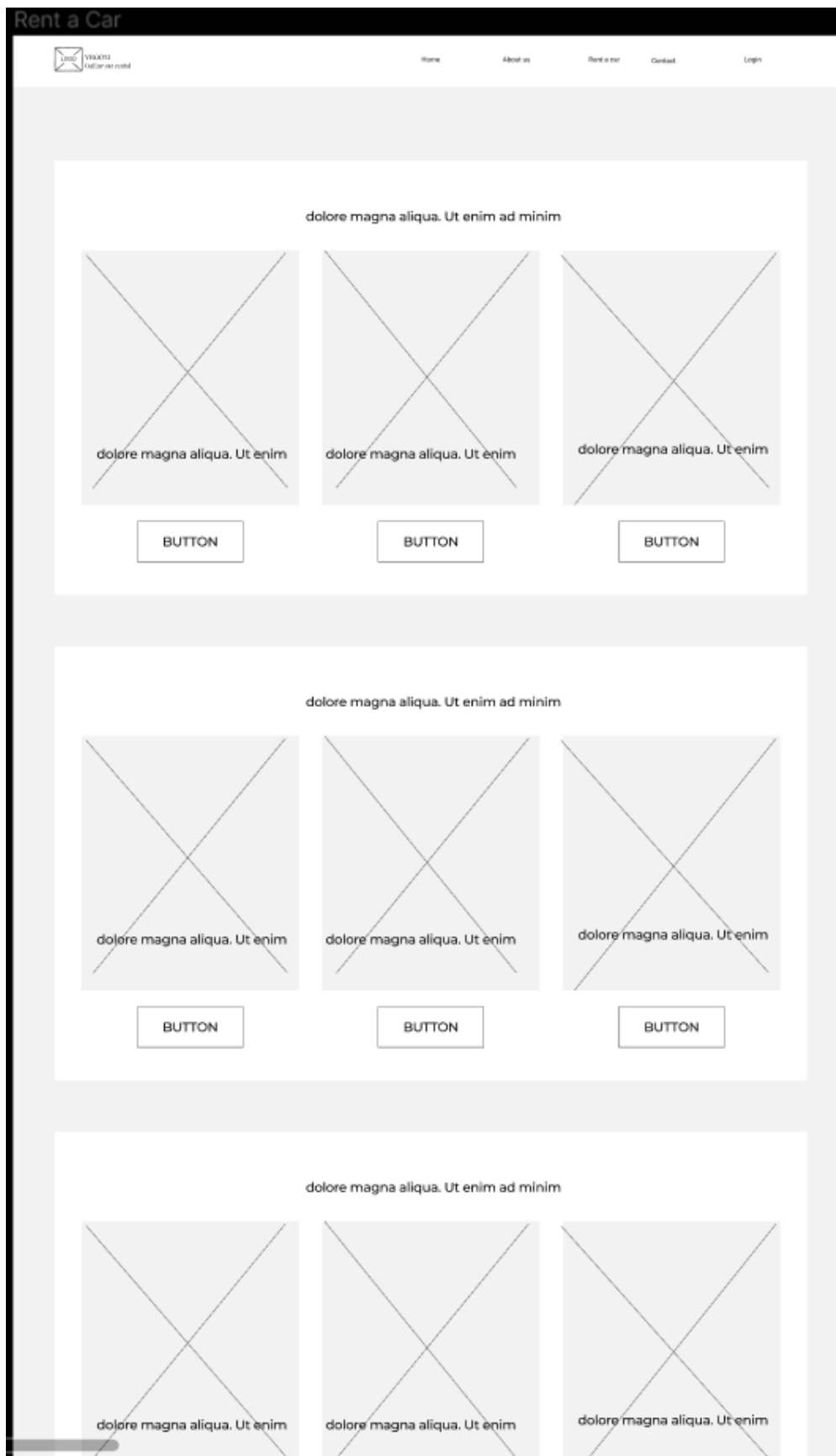


Figure 4 Rent a Car Wireframe

Rent a car pop up

input box / single line

input box / single line

input box / single line

input box / single line

input box / single line

input box / single line

yyyy/mm/dd

yyyy/mm/dd

BUTTON

Figure 5 Rent a Car Pop Up Wireframe

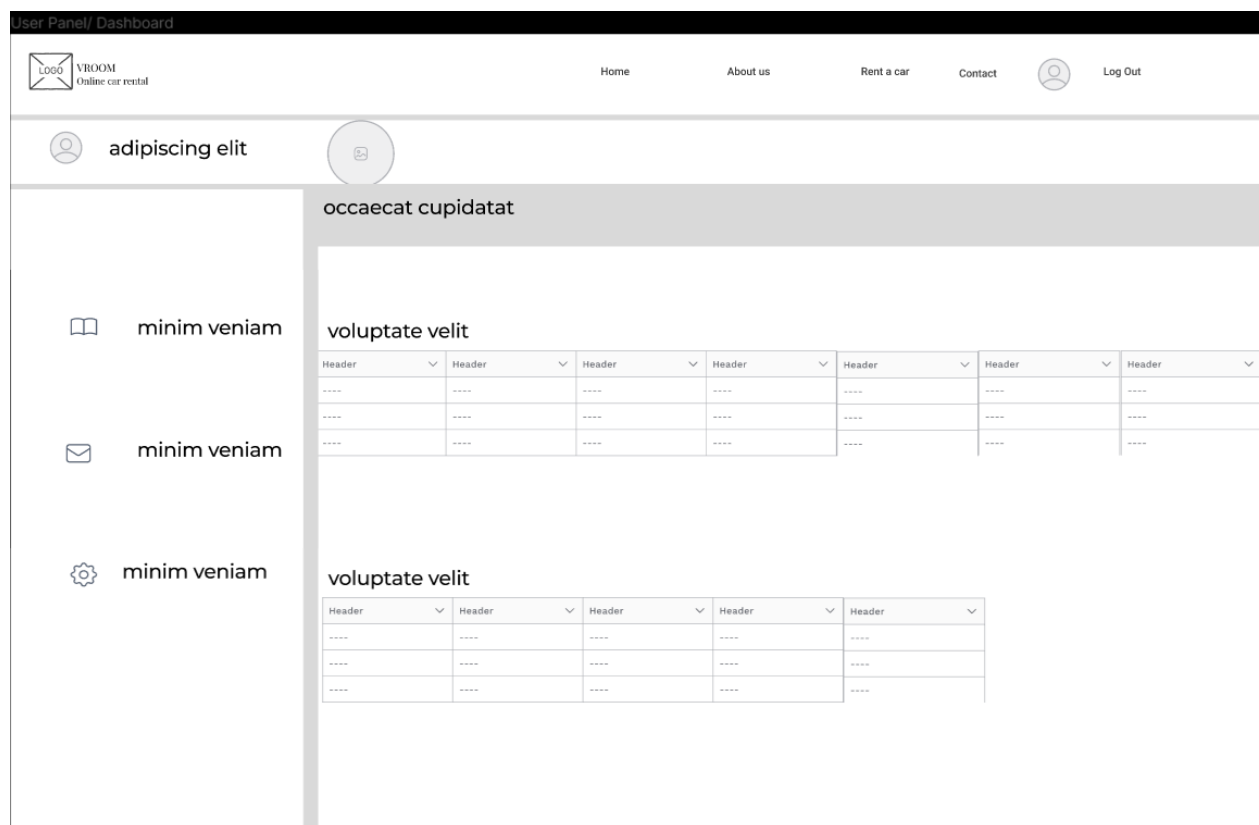


Figure 6 User Panel Wireframe

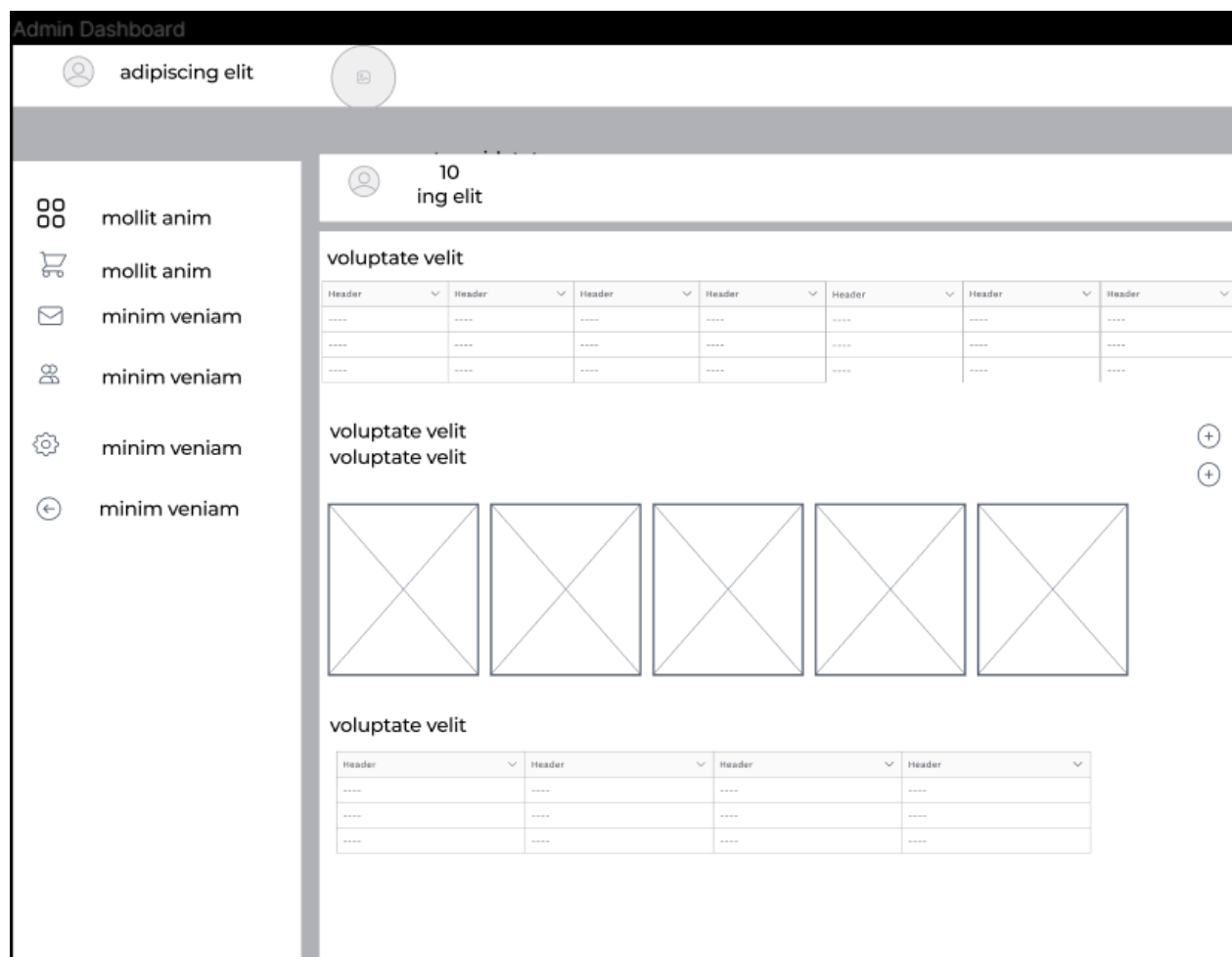


Figure 7 Admin Dashboard Wireframe

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UML Diagrams

Use Case Diagram (Karan Sah : 2329860)

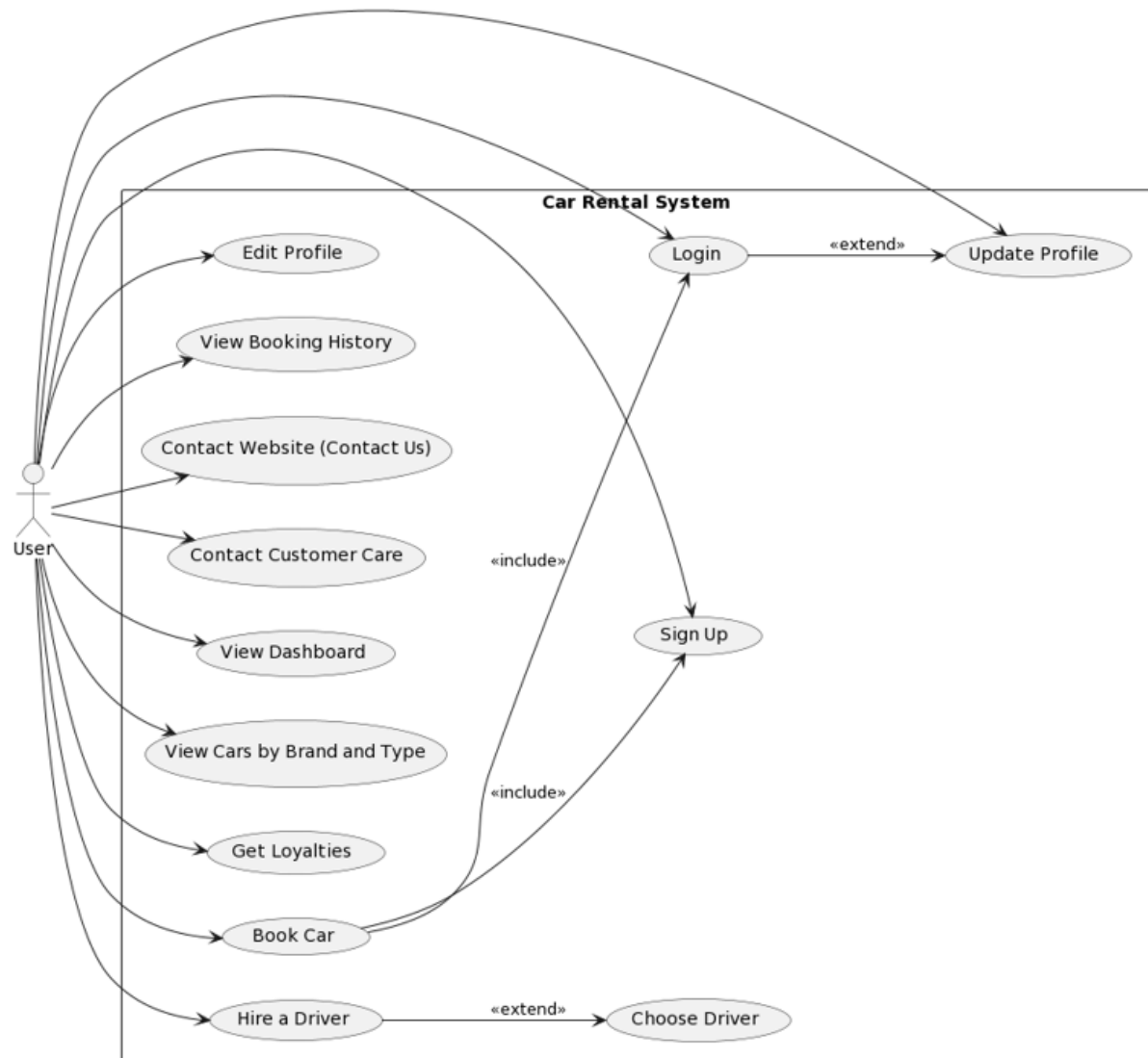


Figure 8 Use Case Diagram (User Side)

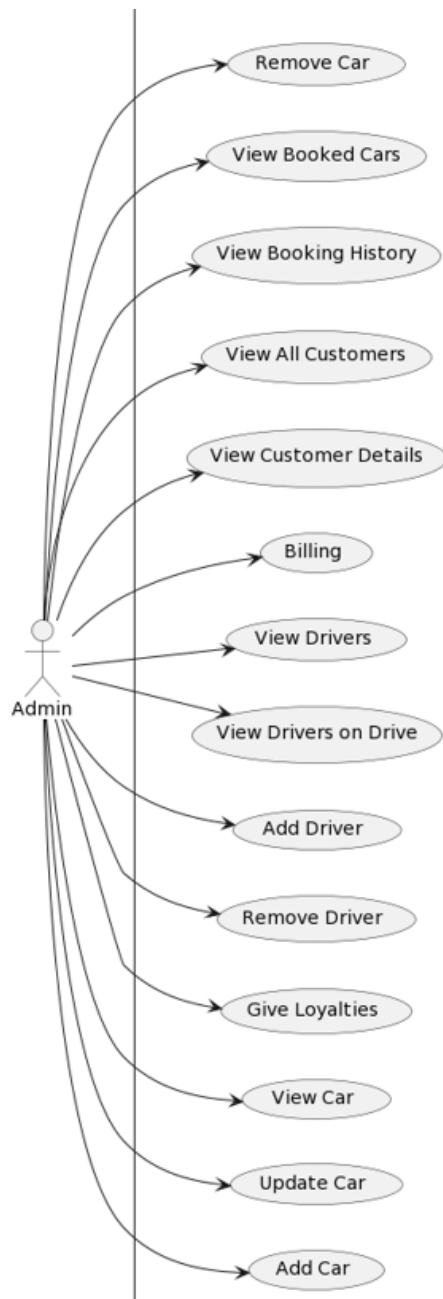


Figure 9 Use Case Diagram (Admin Side)

Activity Diagram for Admin Functions (Karan Sah : 2329860)

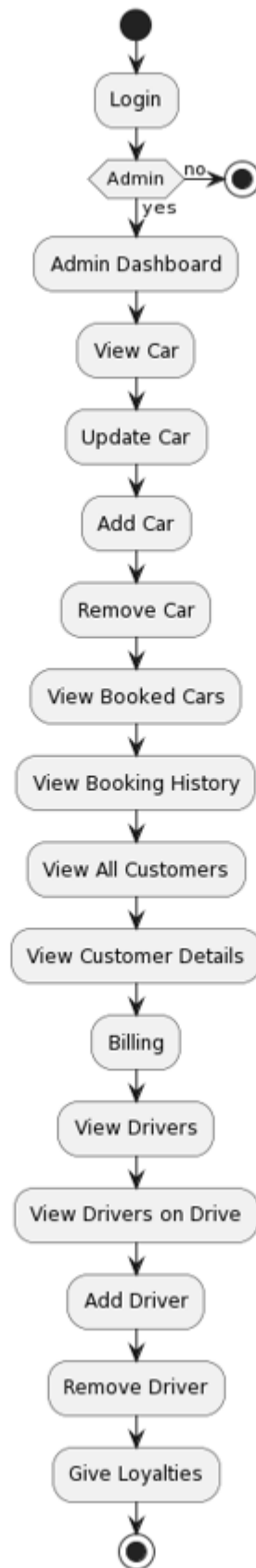


Figure 10 Activity Diagram (Admin)

Activity Diagram for User Functions (Karan Sah : 2329860)

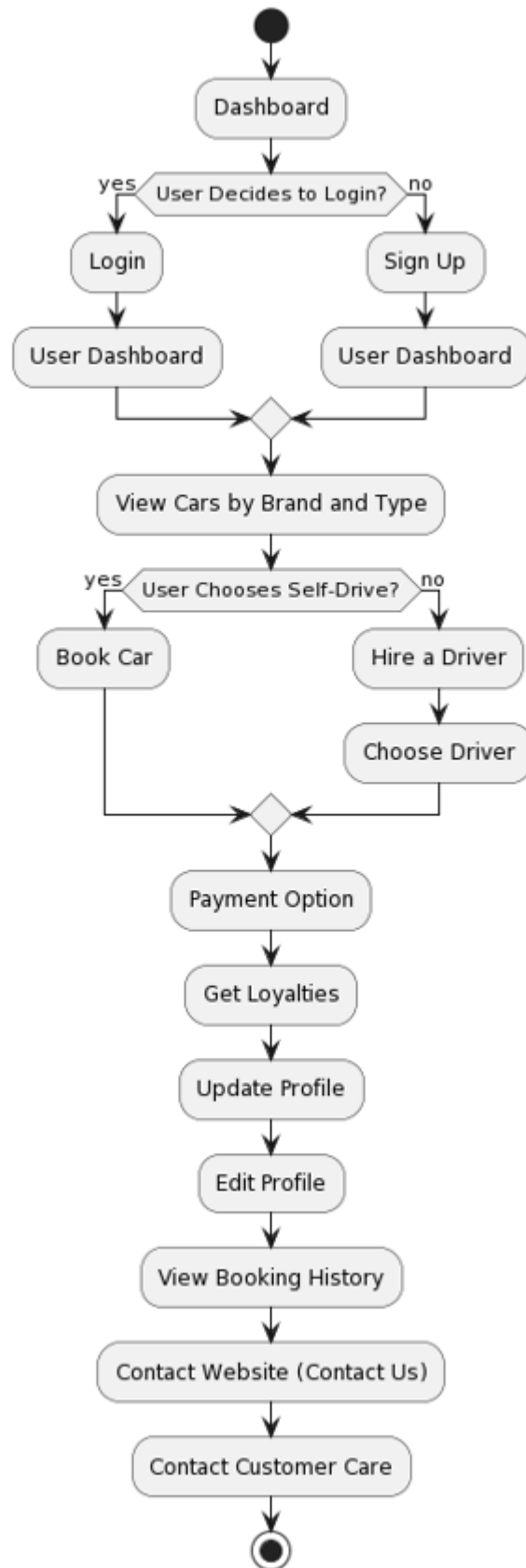
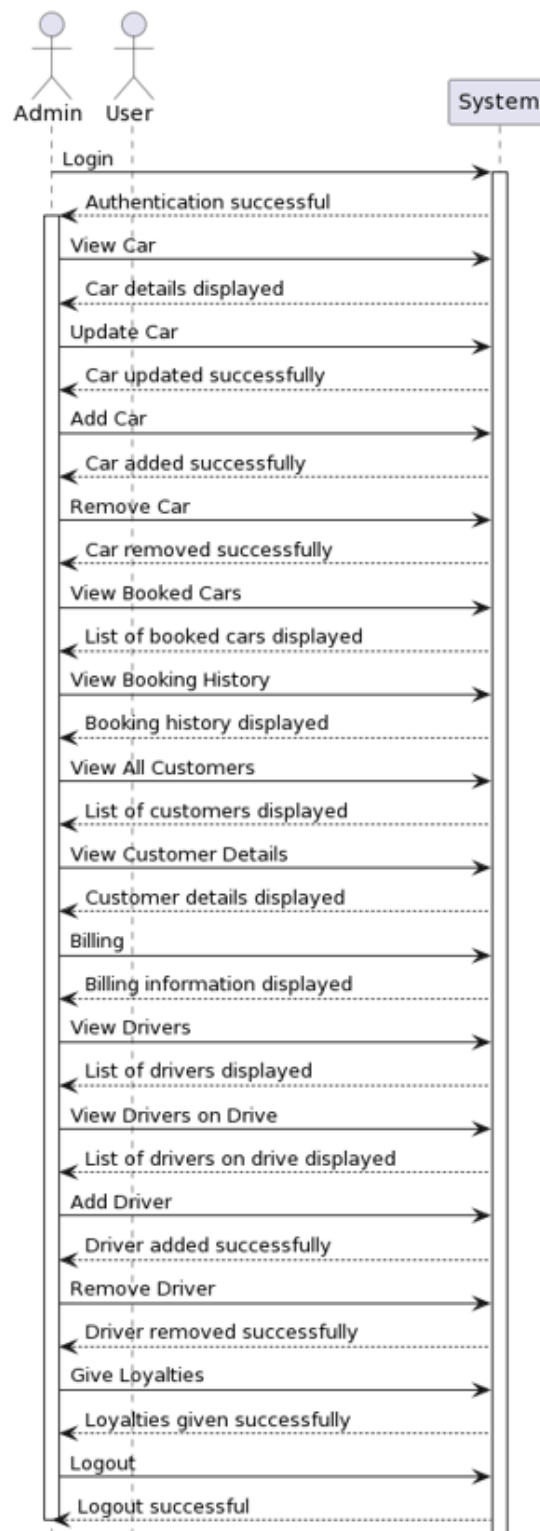


Figure 11 Activity Diagram (User)

Sequence diagram on renting a car / Interaction between Admin and User (Karan Sah : 2329860)



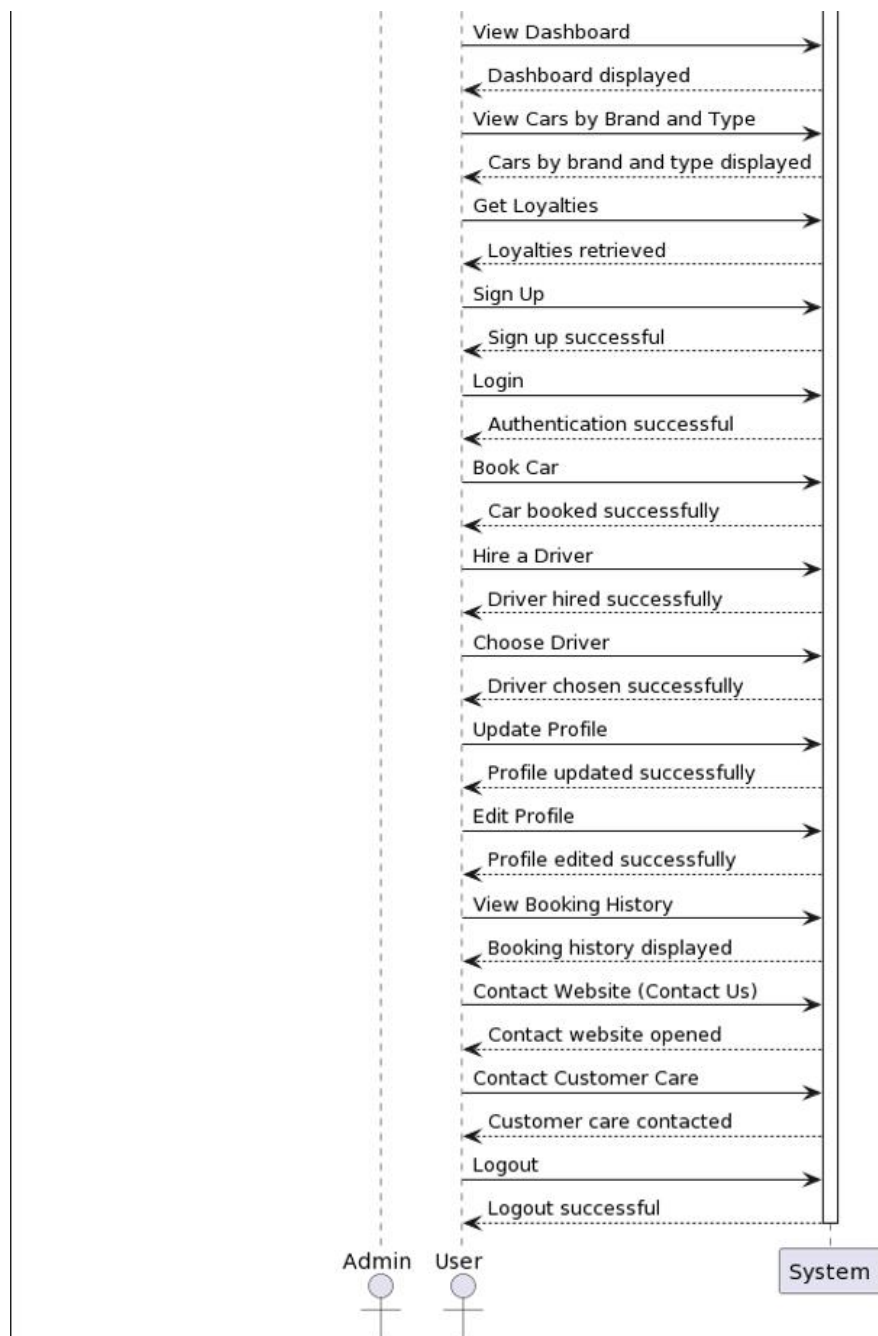


Figure 12 Sequence Diagram (Admin and User)

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ERD

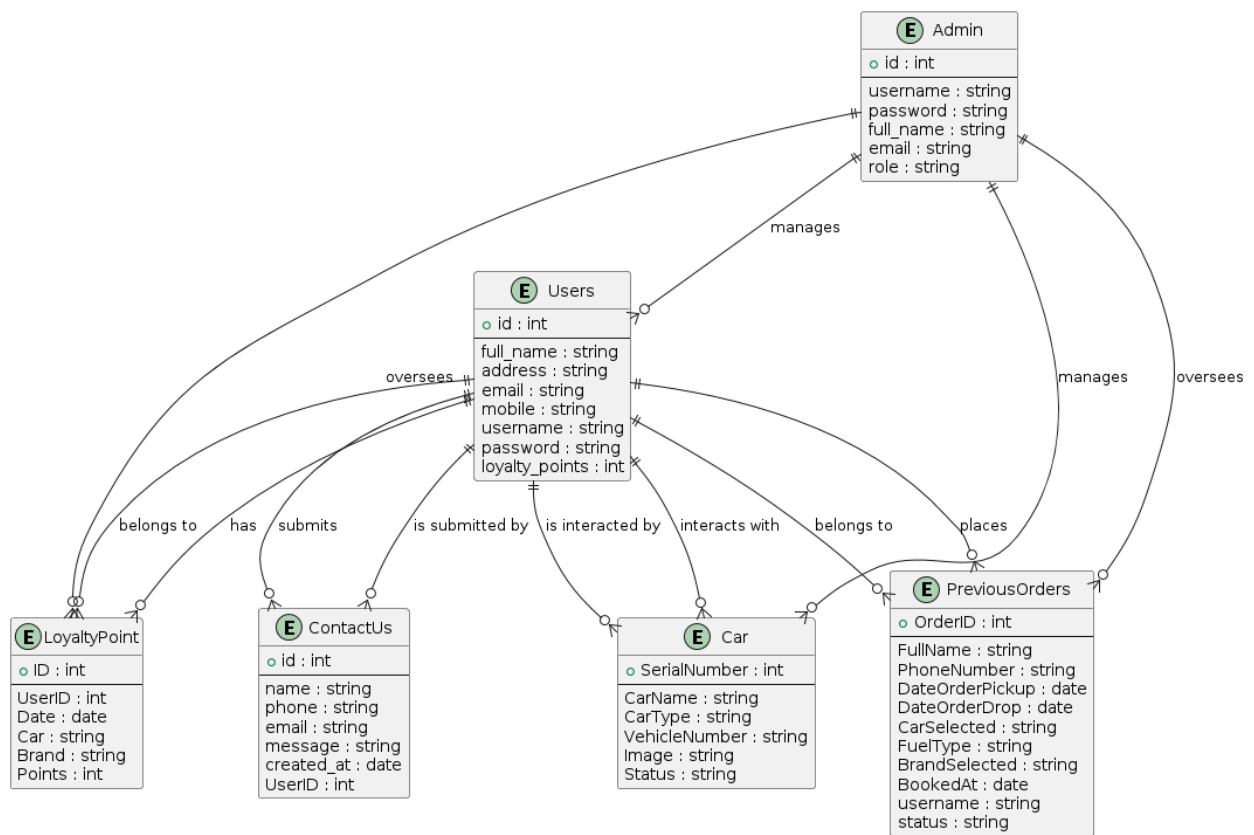


Figure 13 ERD

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Bug Report

Title: CSS of Car Collection of Admin is Mixed and Jumbled

Attachment:

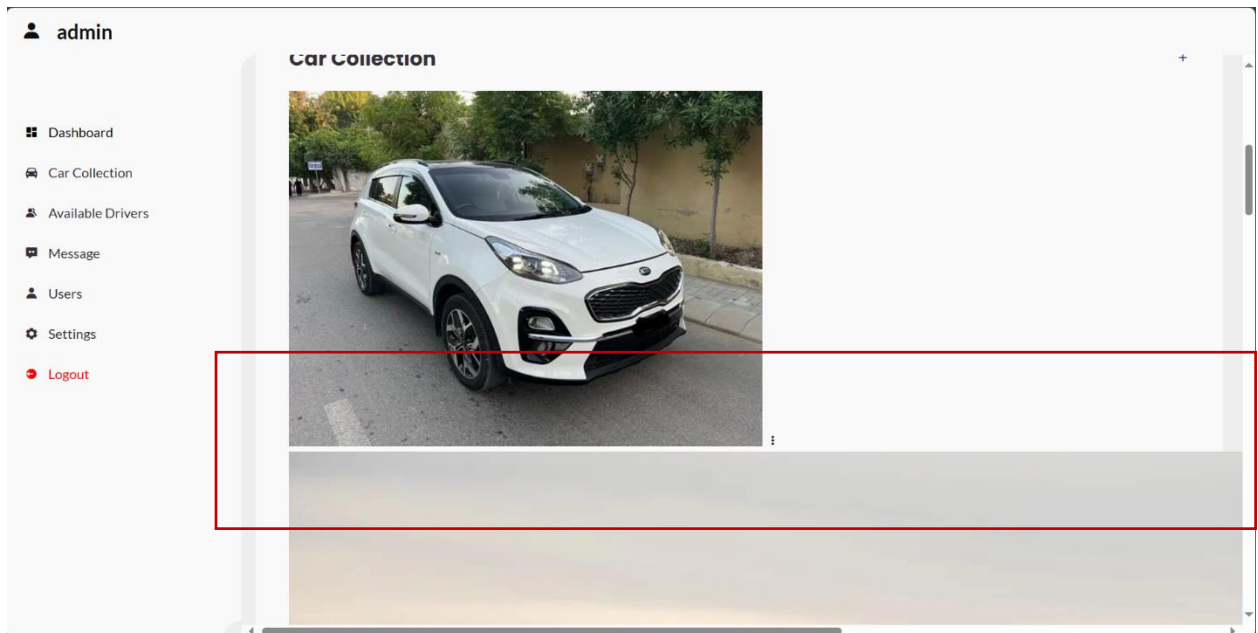


Figure 14 Bug Jumbled CSS

Description: The admin-side automobile collection page's CSS design looks confused, resulting in an uneven and disjointed layout.

Steps to Reproduce:

1. Open the admin panel and log in.
2. Go to the "Car Collection" section.
3. Take note of the haphazard arrangement and design mistakes.

Anticipated Outcome: Every component ought to be arranged and shown in a tidy fashion on the automobile collection page.

Actual Result: Elements are misplaced or overlapped, and the page structure is disorganized.

Title: The Sign-Up Button Is Inactive for Users

Attachment:

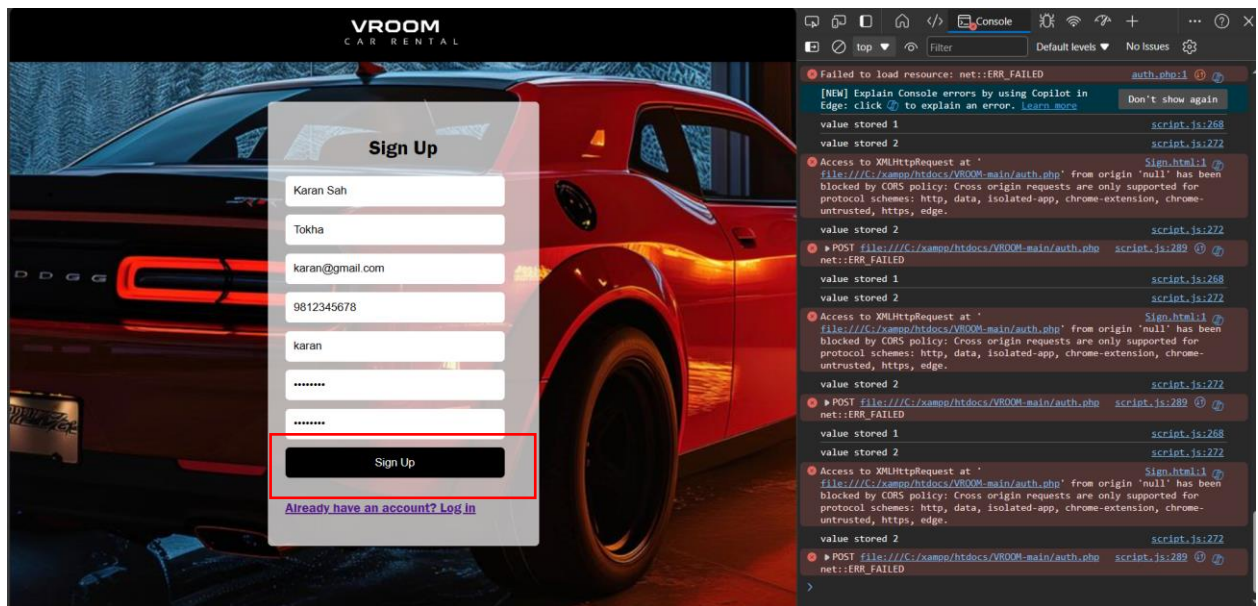


Figure 15 Bug Sign Up Button doesn't Work

Description: Clicking the sign-up button on the user registration page has no effect.

Steps to Reproduce:

1. Go to the page where users register.
2. Complete the fields that are necessary.
3. On the "Sign Up" button, click.
4. On the "Sign Up" button, click.

Anticipated Outcome: The user ought to register and be sent to the relevant page or receive a success message.

Actual Result: Clicking the "Sign Up" button has no effect.

Title: The Driver Add Popup's Exit and Cross Buttons Aren't Working

Attachment:

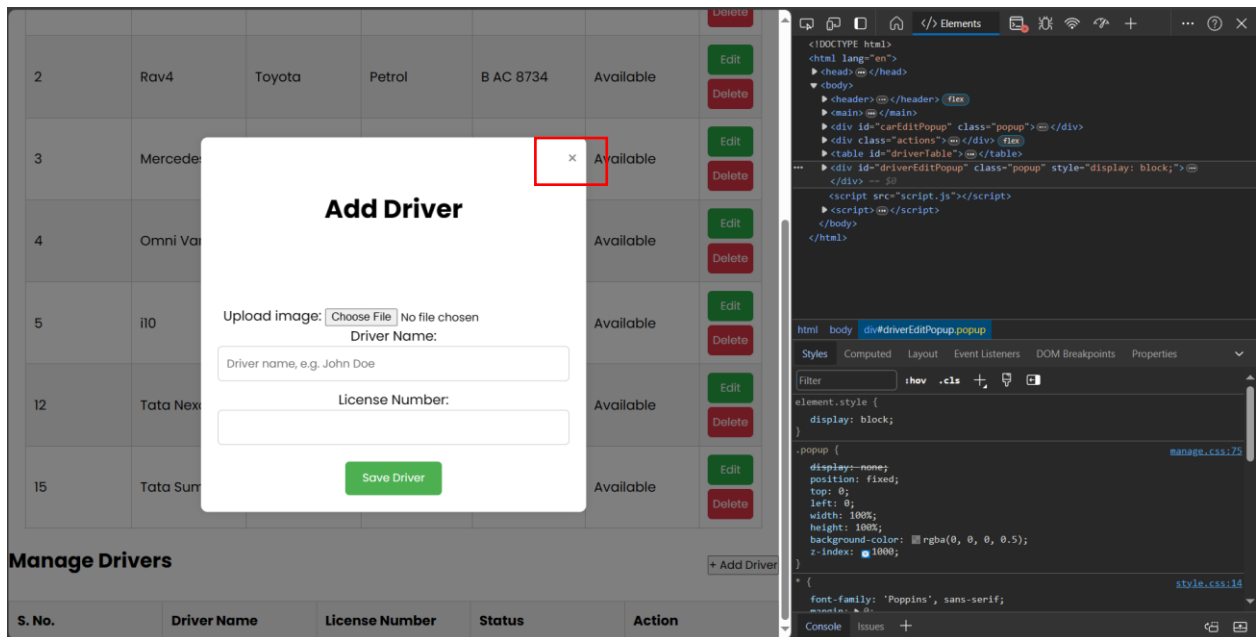


Figure 16 Add Driver - Bug

Description: When clicking the exit/cross button on the "Add Driver" box, the popup window remains open.

Steps to Reproduce:

1. Open the admin panel and log in.
2. Open "Drivers" from the menu.
3. Press "Add Driver" to initiate the pop-up window.
4. Make an attempt to click the exit/cross button to shut the popup.

Anticipated Outcome: When the exit/cross button is clicked, the popup should be dismissed.

Real Outcome: The button does nothing, and the popup window stays open.

Title: Users Aren't Able to Change Their Passwords and It Does Nothing

Attachment:

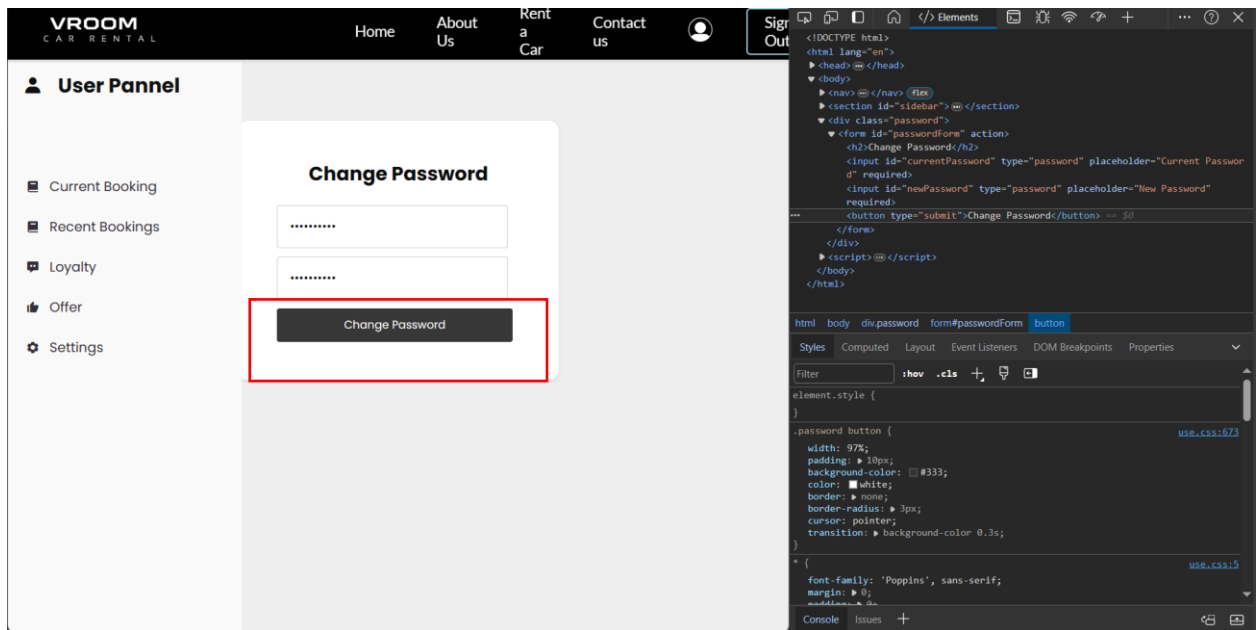


Figure 17 Change Password - Bug Report

Description: There is no way for users to modify their passwords; there is nothing that happens when they click the change password button.

Steps to Reproduce:

1. Open the user account and log in.
2. Go to the section labelled "Change Password" i.e. settings.
3. Enter the new password, the old password, and make sure the new password is correct.
4. Select the "Change Password" button.

Anticipated Outcome: A confirmation message and a changed password should be shown.

Actual Result: Clicking the "Change Password" button has no effect.

Title: User Gets Logged Out When Clicking "Discover Now" Button on Home Page.

Attachment:

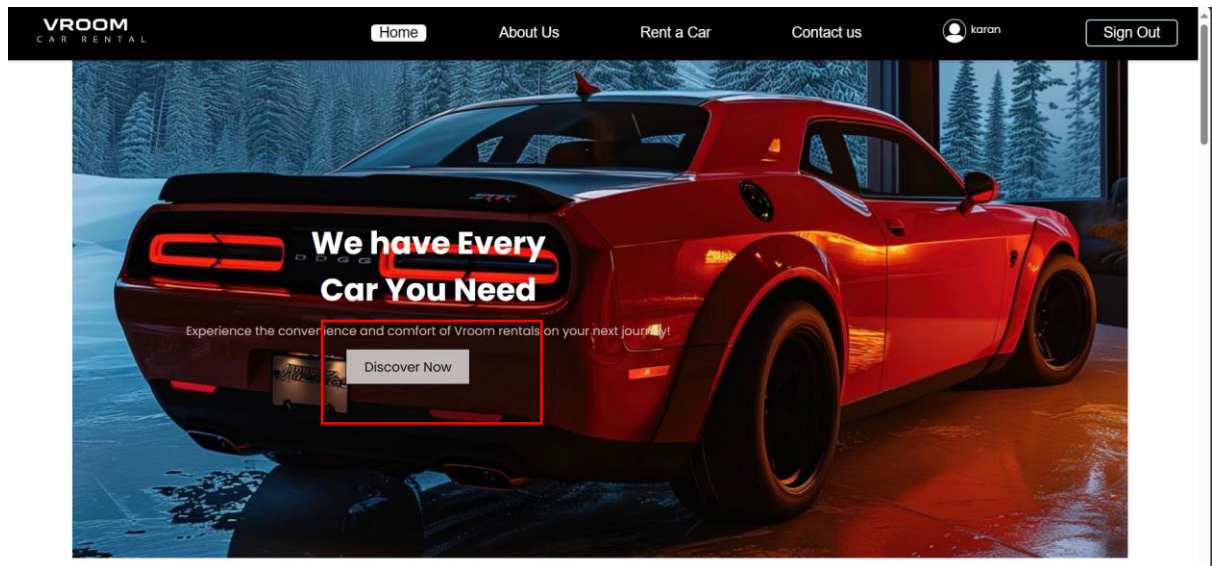


Figure 18 Bug Report - Discover Now

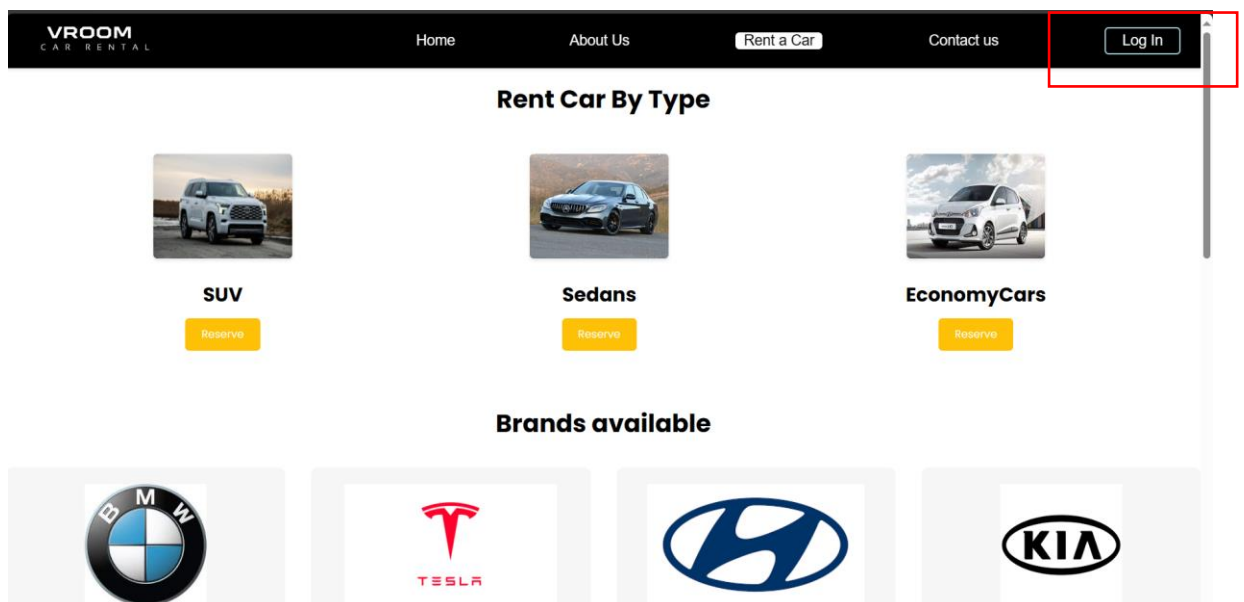


Figure 19 Bug Report - Automatically Log Out

Description: Clicking the “Discover Now” button logs the user out unexpectedly.

Steps to Reproduce:

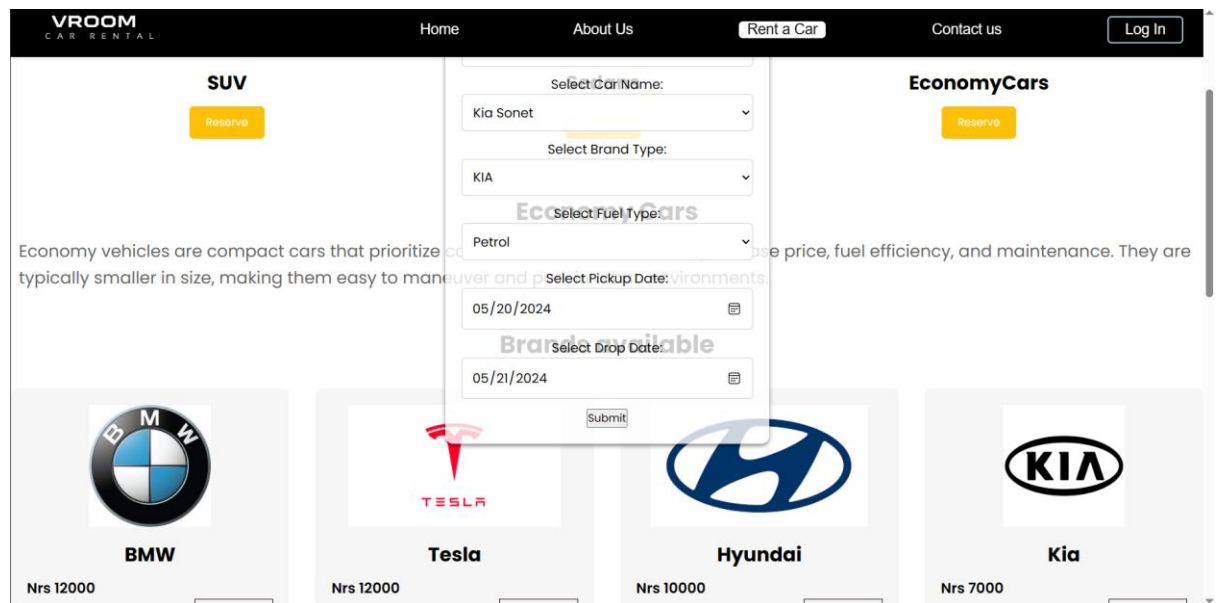
1. Log in to the application.
2. Navigate to the home page.
3. Click on the “Discover Now” button.

Anticipated Outcome: The user should be taken to the discovery page without being logged out.

Real Outcome: The user is automatically logged out and redirected to the login page.

Title: No Confirmation Message Displayed After Booking a Car

Attachment:



The screenshot displays the Vroom Car Rental website interface. At the top, there is a navigation bar with links for Home, About Us, Rent a Car, Contact us, and a Log In button. The main content area is divided into two columns: SUV and Economy Cars. The SUV section features a 'Reserve' button and a description of SUVs. The Economy Cars section features a 'Reserve' button and a description of economy vehicles. A central booking form is overlaid on the page, containing fields for Select Car Name (Kia Sonet), Select Brand Type (KIA), Select Fuel Type (Petrol), Select Pickup Date (05/20/2024), and Select Drop Date (05/21/2024), with a Submit button. Below the form, there are four car options displayed as cards: BMW (Nrs 12000), Tesla (Nrs 12000), Hyundai (Nrs 10000), and Kia (Nrs 7000). Each card includes the car's logo and a 'Reserve' button.

Figure 20 Bug Report: No Confirmation Message

Steps to reproduce: No success or confirmation message is shown after booking a car.

Steps to Reproduce:

1. Log in to Vroom.
2. Go to the car booking section.
3. Fill the sections required.
4. Click on the "Submit" button.

Anticipated Outcome: A confirm message is shown after booking the car.

Real Outcome: No message is displayed, which makes the user uncertain.

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Test Cases

Functional Testing:

TC ID	Project/Module	Req ID	Scenario	Test Case Title	Steps	Test Data	Expected Results	Actual Result
1	User Signup	U4	Signup	Verify Signup	1. Navigate to the signup page. 2. Enter user details 3. Click the "Sign Up" button	Name: Karan Sah Address: Tokha Email: karan@gmail.com Phone number: 9812345678 User Name: karan Password: karan12345 Verify Password: 12345Karan	1. User should be able to Navigate to the signup page. 2. User should be able to enter user details 3. User account created successfully	1. Navigate to the signup page. 2. Enter user details 3. Noting happens
2	Admin Management	A12	Add Driver	Verify Driver Addition	1. Access admin dashboard section 2. Navigate to "Add Driver" 3. Enter driver details 4. Click "Add" button	Upload Image: Uploaded Imag Driver Name: Dennis Katuwal License Number: 12-16-1789	1. Admin should be able to access admin dashboard section 2. Admin should be able to navigate to "Add Driver" 3. Admin should be able to enter driver details 4. Driver added successfully	1. Access admin dashboard section 2. Navigate to "Add Driver" 3. Enter driver details 4. Driver not displayed in frontend table Cross button inactive
3	User Account Management	U9	Change Password	Verify Password Change	1. Log in to user account 2. Navigate to account settings 3. Click on "Change Password" option 4. Enter current password and new password 5. Click "Change Password" button	Current password: karan12345 New password: 12345karan	1. User should be able to log in to user account 2. User should be able to navigate to account settings 3. User should be able to click on "Change Password" option 4. User should be able to enter current password and new password 5. The password changes and shows confirmation message.	1. Log in to user account 2. Navigate to account settings 3. Click on "Change Password" option 4. Enter current password and new password 5. The button doesn't work and nothing happens
4	Website UI	A1	CSS Issue	Verify Car Collection CS	1. Navigate to the car collection page. 2. Observe the layout and styling	N/A	Admin should be able to see car collection CSS properly aligned and styled	Car collection CSS jumbled and affecting website appearance
5	Booking System	A2, A3	Column Data Mix	Verify Data in Columns	1. Navigate to the bookings page. 2. Check the "Price per Hour" Column 3. Check the "Status" Column 4. Check the "Action" Column	N/A	"Price per Hour" should show numerical values "Status" should show status indicators "Action" should have action buttons	"Price per Hour" column shows status data "Status" column shows action buttons Action column is empty

Status	Comments(Product Owner)	Type of Testing
Passed		Functionality
Passed		
Failed	User signup functionality not working as expected. Please investigate and fix.	
Passed		Functionality
Passed		
Passed		
Failed	Drivers should be visible in frontend table after addition. Please investigate why they are not appearing and why the cross button is inactive.	
Passed		Functionality
Passed		
Passed		
Failed	Users should be able to change their passwords. Please investigate why the functionality is not working as expected.	
Failed	Css of car collection need to be fixed to improve website view. Please investigate and rectify the issue.	Functionality
Passed		Functionality
Failed	Columns display incorrect data.	
Failed	"Price per Hour" should display prices	
Failed	"Status" should display status indicators	
	"Action" should display action buttons.	
	Investigate and correct the data mapping.	

Figure 21 Functional Testing

Unit Testing

6	Car Rental Service	U6	Rent a Car	Verify Car Renting Proo	1. Navigate to the "Rent a Car" on the navigi Name: Karan Sah 2. Select a car type (SUVs, Sedans, Economy 3. Click on the "Reserve" button. 4. Enter details.	Email: karan@gmail.com Phone Number: 9812345678 Car Type: KIA Sonet Brand Type: KIA Fuel Type: Petrol Pick Up Date: 2024-05-18 Drop Date: 2024-05-21	Renting process should complete with data on database.	Renting process completed with information in database.
7	Admin Management	A10, A1	Add New Driver	Verify adding new driv	1. Login as admin. 2. Navigate to "Available Drivers". 3. Go to "Manage Drivers." 4. Click on "Add Driver" button 5. Upload a picture 6. Fill in the details. 7. Click "Save" button.	Username: admin Password: admin123 Driver Name: Roshan Chapai License Number: 012345-2525 Phone Number: 9812345678 Address: Ganganabu	New driver is saved successfully and displayed in the driver list	Driver is saved successfully and displayed in the driver list
8	Car Management	A1, A2,	Add new car	Verify adding new car	1. Login as admin 2. Go to "Car Collection" 3. Navigate to "Manage Cars" 4. Click on "Add Car" button. 5. Upload an image. 6. Fill in the details 7. Click "Save Car" button	Car Name: Ford 1957 Car Brand: Ford Car Type: Petrol Price per Hour: Rs. 1000 Vehicle Number: BA 3 KHA 2022	New car is saved successfully and displayed in the car list	Car is saved successfully and displayed in the car list

Passed	Car renting functionality works as expected.	Unit
Passed	Driver addition functionality works as expected	Unit
Passed	Car addition functionality works as expected.	Unit

Figure 22 Unit Testing

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Screenshots of GitHub Issues

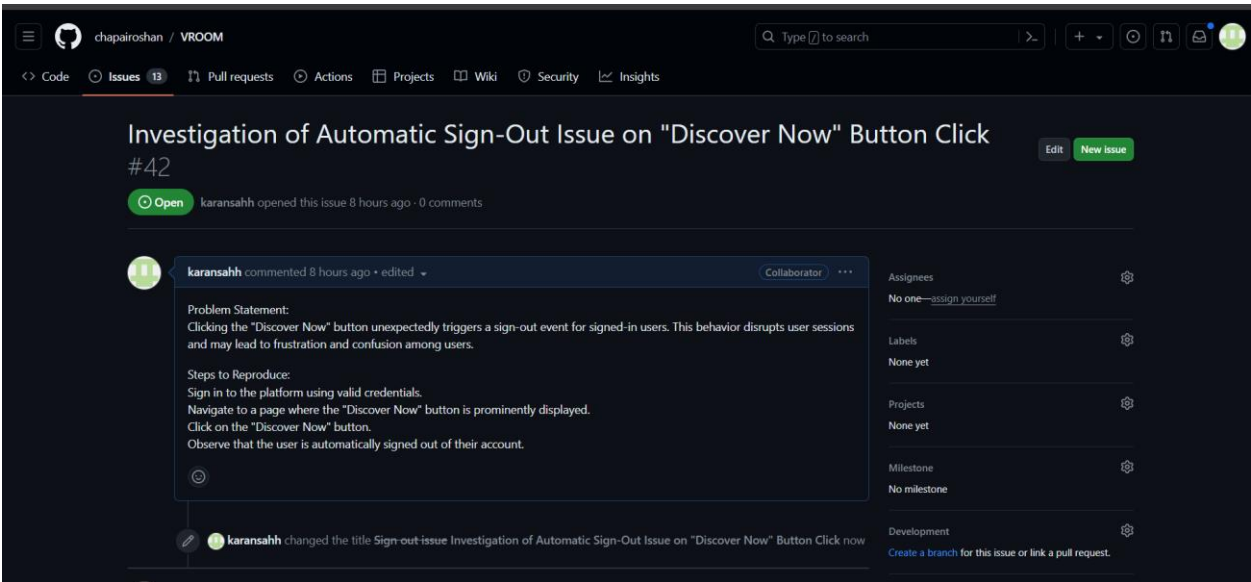


Figure 23 GitHub Issue Automatic Sign Out

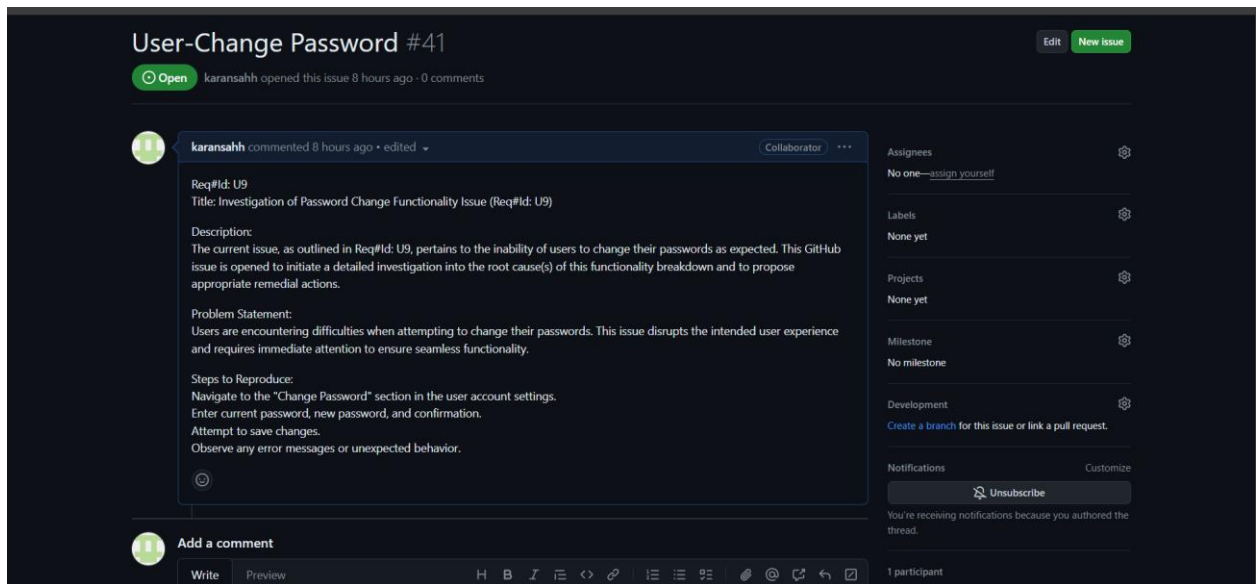


Figure 24 GitHub Issues User Can't Change Password

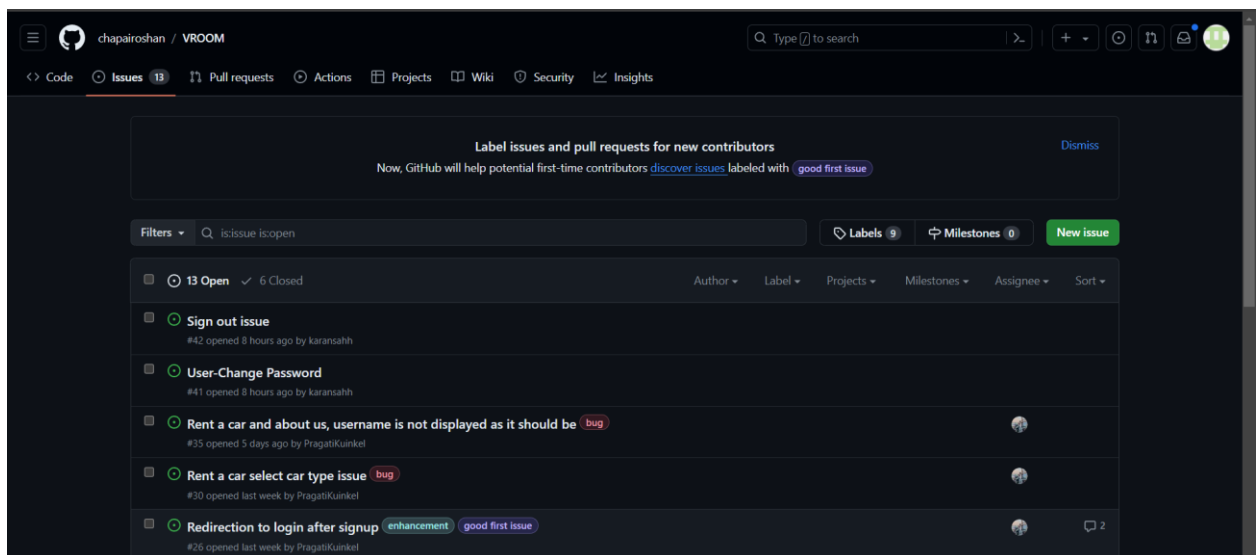


Figure 25 All GitHub Issues

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Appendix B
Collaboration Document
Good Communication
Offline Meetings



Figure 26 Offline Meeting

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Google Chats

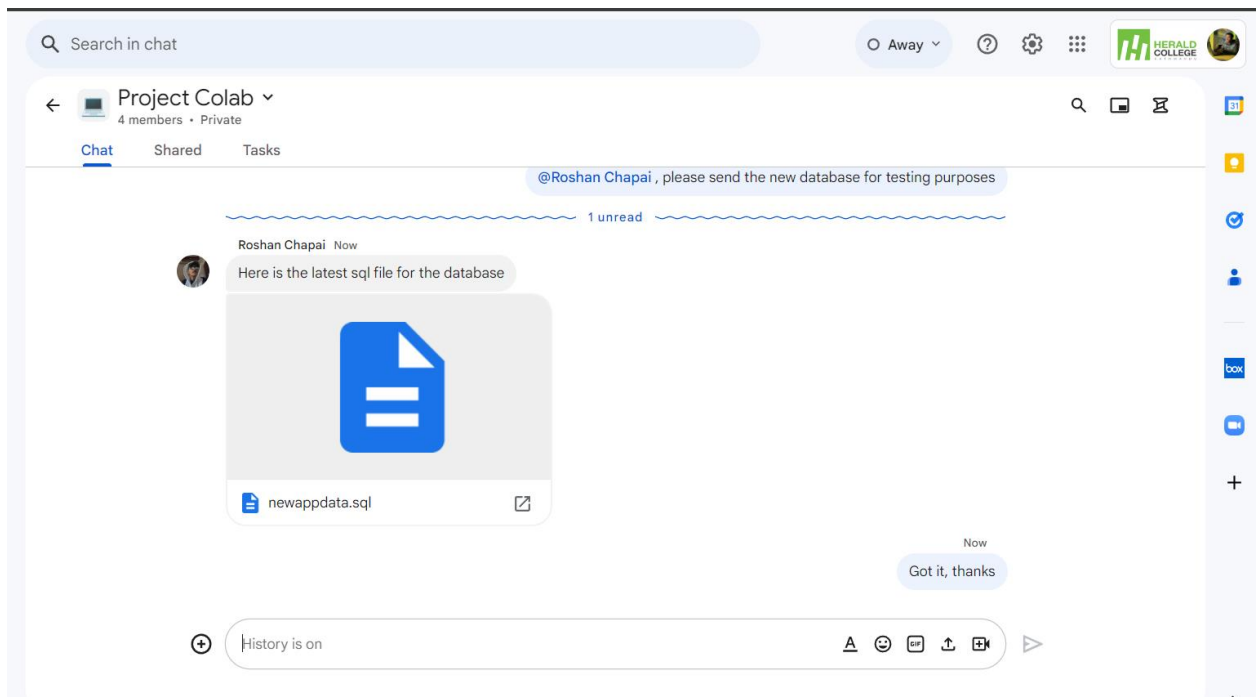


Figure 27 Google Chats

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Files Shared through Jira

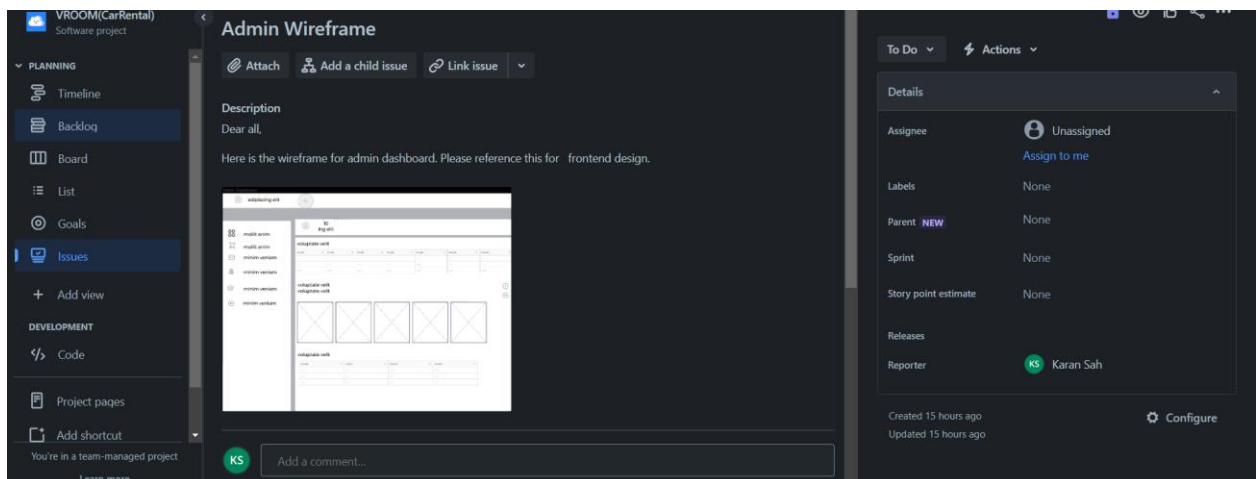


Figure 28 Admin Wireframe Shared through Jira

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Online Meetings

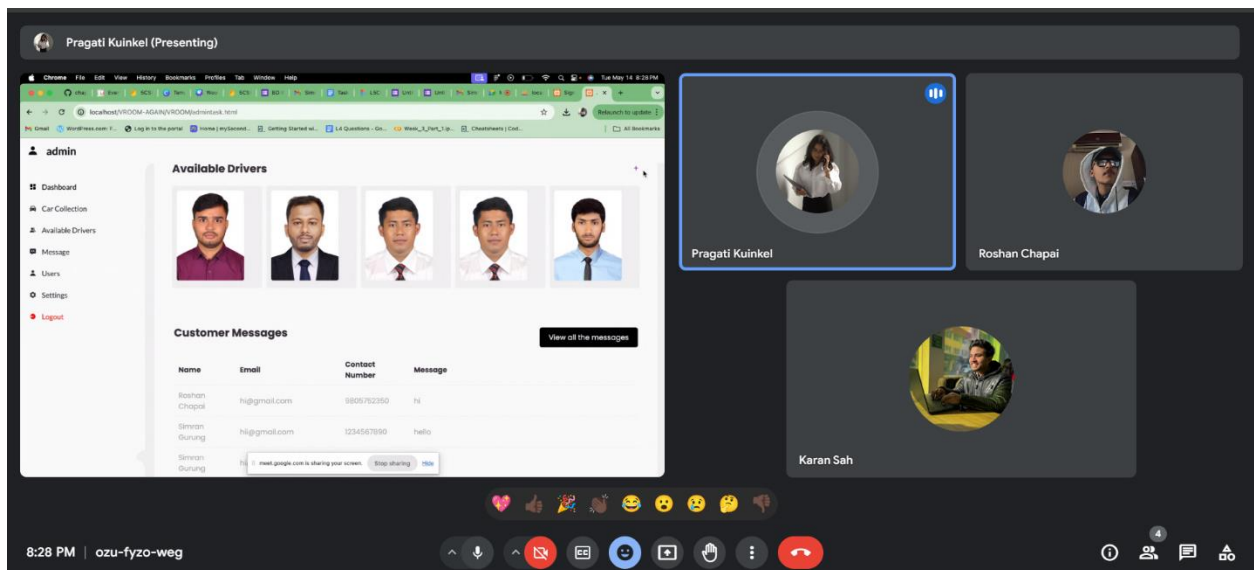


Figure 29 Online Meetings

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Continuous Personal Development Report

Personal development is the commitment to oneself for lifelong learning, refreshing the skills and improving them, learning new one, retaining the known knowledges, and keeping up to date with the industry. There are vivid courses taking, and learning new things through different mediums (The CPD Certification Service, 2020).

Though Sprint 1 had a significant impact on my personal development during the ending process working on deadlines during Sprint 2 had an even greater impact on my growth. The technical, and professional aspects had a rapid rise due to all the processes like handling conflicts that arose during the project completion, working on deadlines due to sprint, and handling client meetings. The major element for improvement had to be how to work on a team thus increasing my ability and capability to work in a team. The real-world development team majorly has to work in a team thus making it easier and as a training for the real world collaborations too. In this case, the development was done when we had meetings and things to discuss. Making agendas in meeting minutes and completing them was a kind of accomplishment. Likewise, the conflict management was getting out of hand and after watching YouTube videos and gaining insight from that the conflict were handled easily.

Similarly, the games that were played during the tutorial and workshop classes were a lot helpful for conflict management and teamwork.

CPD is to know and make a plan, learn the said and related works and skills, reflect after the learning on learned materials, apply the knowledge learned, and share the knowledge. It goes as in cycle and is personal thus contributing immensely to personal growth and development (Skills You Need, 2011). Thus according to this when reflected I had gained a lot from this module in terms of growth. The growth was through self-research on unfamiliar topics, how business analysts solve their roles, their responsibilities, and the workings and ins and outs of collaborative development. This was a great help in many terms such as working in real life environment too. Though self-researching had helped and aided immensely in this growth, nevertheless implementing those learned cases and working with other team members with different ideas was surprisingly better for this development.

On reflecting on the whole journey of Sprint 2 as well as Sprint 1, although it was filled with deadlines, conflicts, hectic implementations, misunderstanding; the whole journey nevertheless has contributed immensely to my growth. The experience was definitely a test for my patience, resilience, and my ability to adapt which not only I overcame but also learned a lot. The project of the car rental system increased my capability to work in a team, make team-based decisions, and my overall experience in handling pressures.

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YouTube for Conflict Management:

<https://www.youtube.com/watch?v=D--X9-726bk>

YouTube

for

Business

Analysts:

<https://www.youtube.com/@karaleise>

Meeting Minutes:

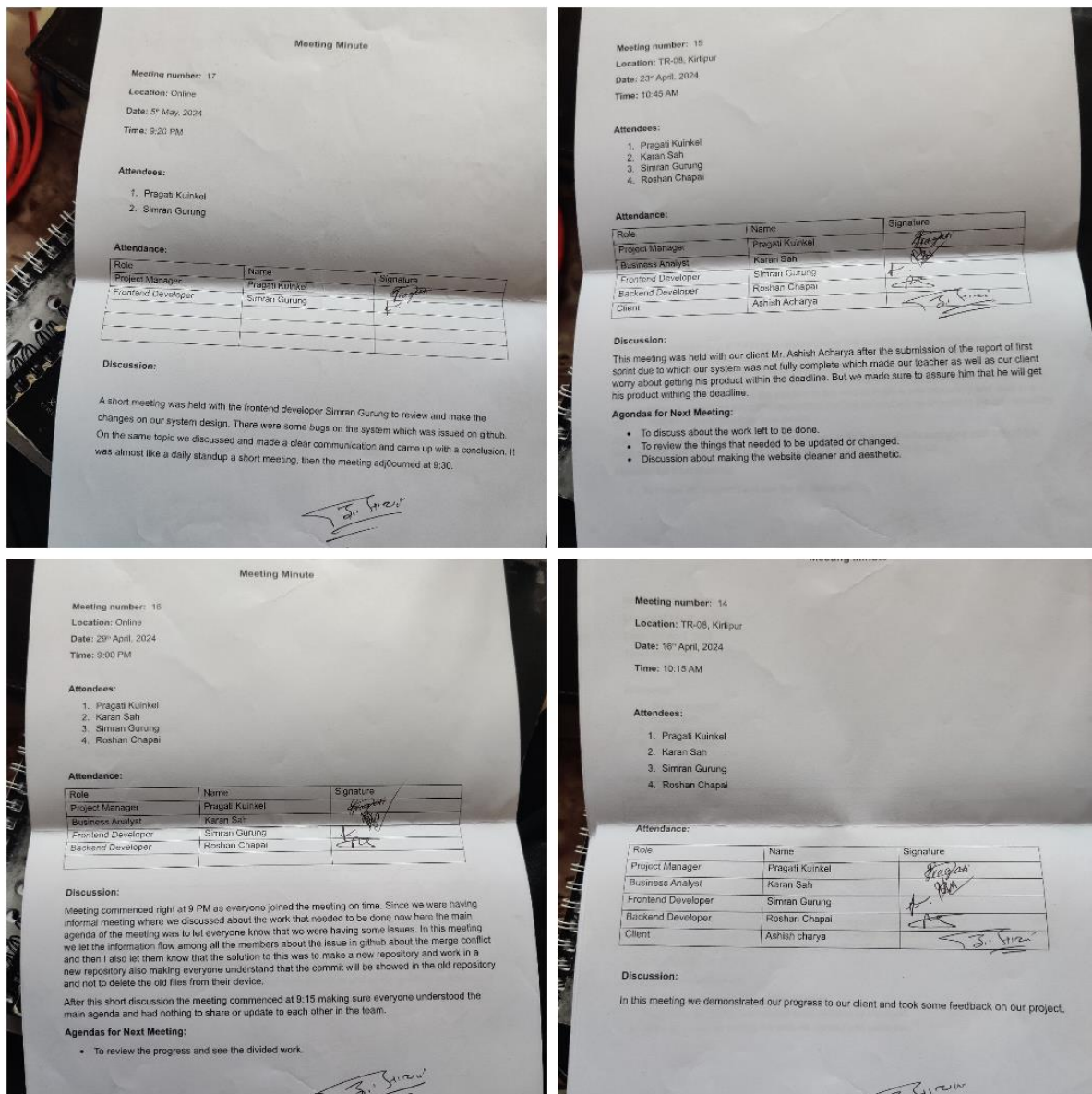


Figure 30 Meeting Minutes

YouTube for Collaborative Development

<https://www.youtube.com/@5MinutesEngineering>

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Issue Tracking

Report on Issue Tracking

The tracking of issues was helpful to the overall development of the project whereby timely tracking and reporting of the issues were helpful in solving the issues timely thus stopping the building of the website on top of errors and mistakes. The tracking was done through regular checking of the works and development. The issues were then listed on GitHub and Jira so that other team members and I could track the issues and solve the issues that occurred during the development process. The identified issues were then on proceeded to solve through online and offline meetings and the solutions were discussed through team meetings and collaborations.

The queries, questions, and complaints are listed. These are the types that are raised by users during functional and unit testing. Customers or users reach out to report such issues and aid in gathering feedback for relevant breakthroughs. Effective tracking of issues is helpful for prioritizing the resolutions and issues (servicenow, 2024).

The issues that occurred were simple problems whereby first issue was when the user wasn't able to change their password. The Req#Id of U9, whereby the investigation was launched for this functionality to propose remedies. Likewise, by clicking on Discover Now on the homepage after the user is logged in, the user gets automatically logged out whereby the investigation was launched for the possible solutions. Similarly, the sign-up button when users enter their details wasn't working. There were different bugs, enhancement required as well as help required in issues whereby they all were appropriately solved through teamwork.

The issues are solved first by identifying the actual problem, the person who has been experiencing the problem, the starting point of the problem, the problem occurring is whether intermittent or constant, are the changes occurring recently or they have been changed before, would the problem be able to recreate, are the logs of the issues available. Thus, first identifying and finding questions to these issues would be the first step in solving the issues that occurred (Source One, 2024).

Thus these were the steps taken during the tracking and solving the issues. The effective tracking of issues made it possible for timely solving of the issues which in turn saved us a lot of frustration and times. The thing most learned from the process

of issue tracking is how crucial it is for any project as the timely finding of any error is very crucial in any project which in turn saves the resources required for the project. The issue tracking ensures the project is up to the standard and is of quality at any time of the time. This is for the smooth operation of the website and for user satisfaction. It was very helpful in understanding the concept of issue tracking, how the issues are tracked, and how timely it is informed to the other members. Issue tracking ensures that the end users who are user and admin can go to the complex process of development and deliver higher quality of websites.

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Users Can't Change their Password Issue in GitHub

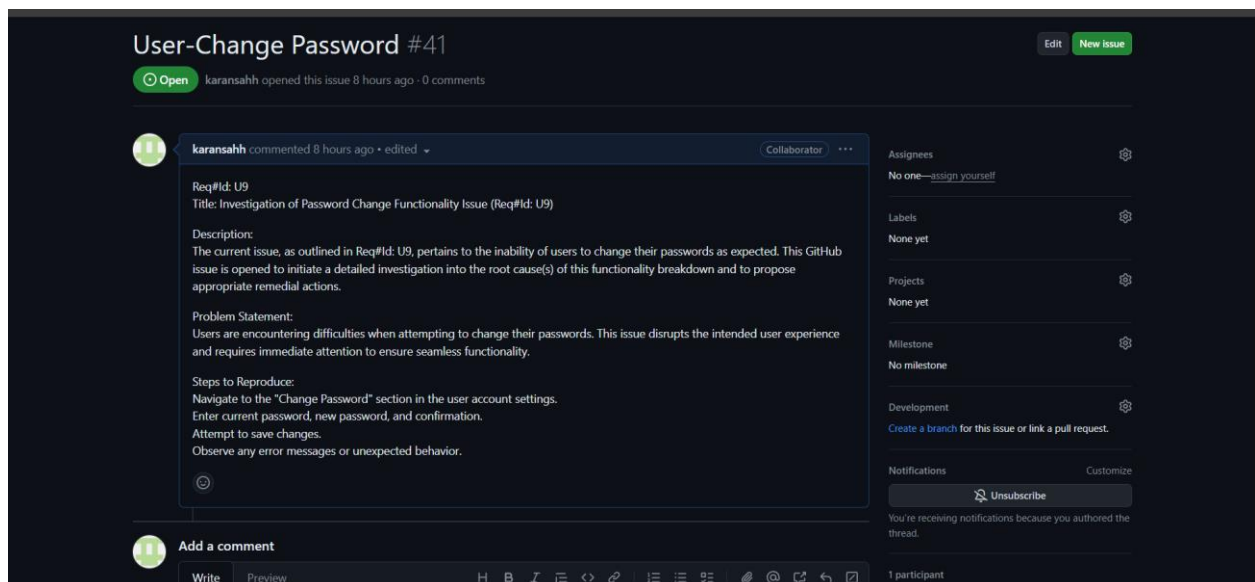


Figure 31 Issue - Password can't be changed

Automatic Sign out User Issue in GitHub

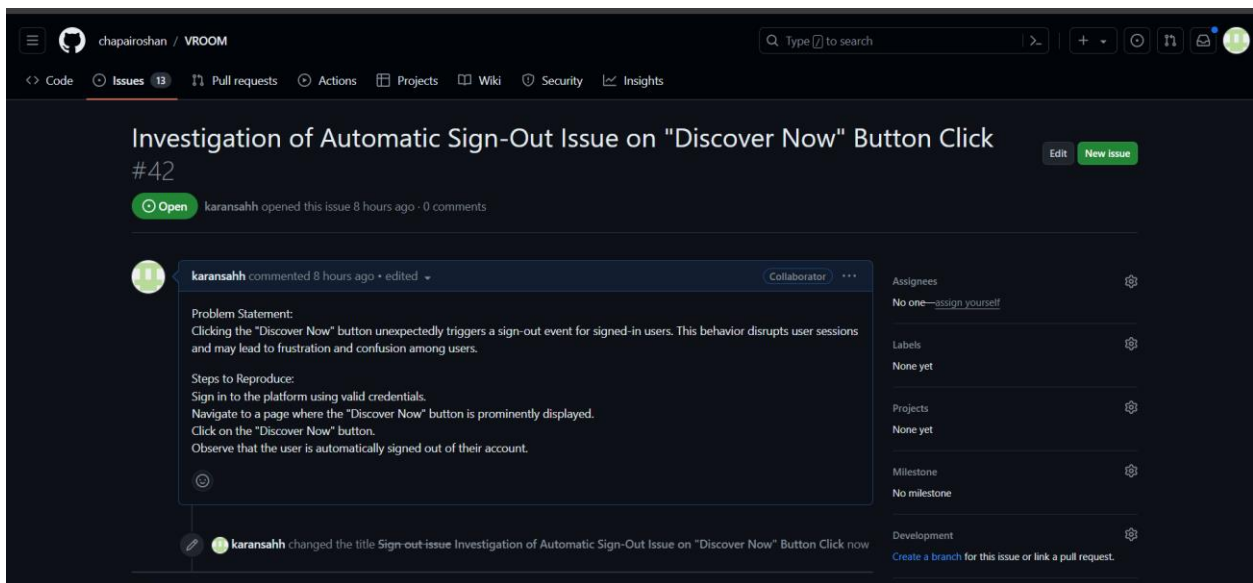
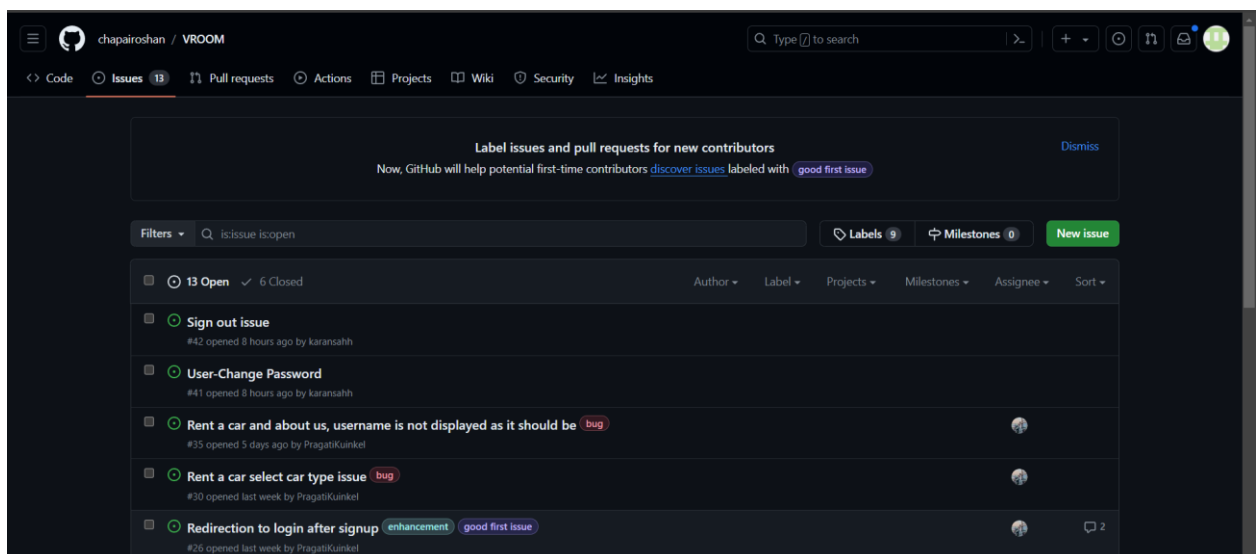


Figure 32 Issue - Automatic Sign Out

Overall Issues in GitHub



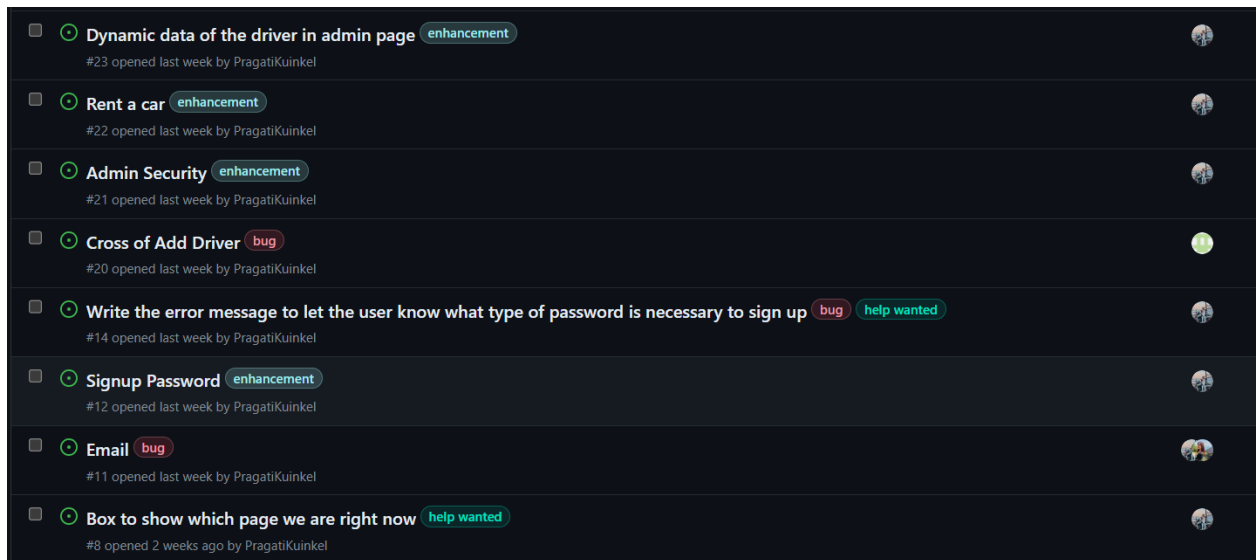


Figure 33 Issue - All Issues in GitHub

Sign Up Button Not Working Issue in Jira

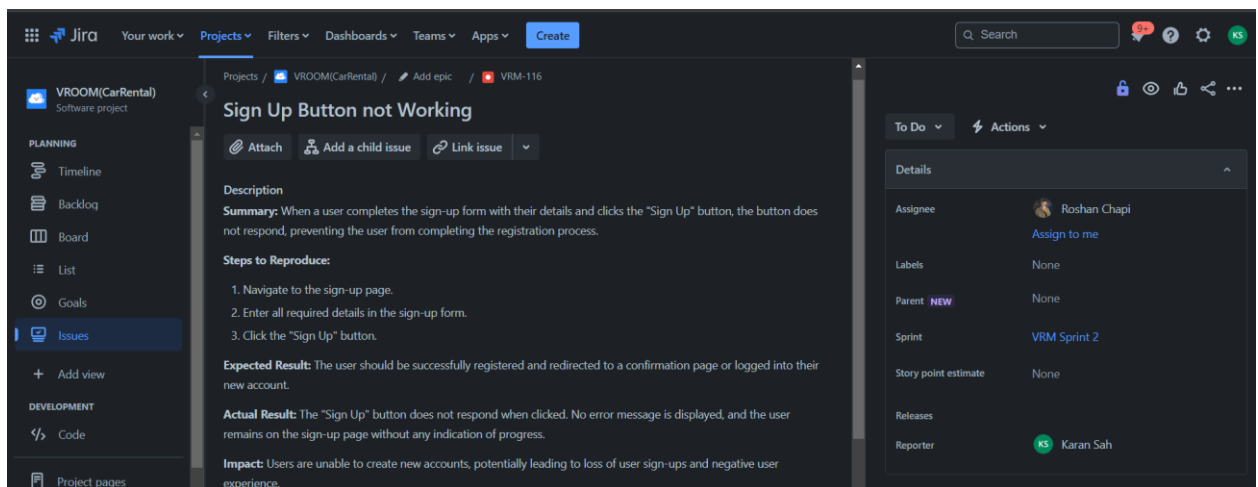


Figure 34 Issue - Sign up Button not working in Jira

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Sprint Retrospective

Every member played a very crucial role in completing the project. Most of the user stories were completed. Despite having less time to complete, all the significant user stories were done and submitted. Especially in sprint 2, there were almost double the tasks compared to sprint 1 which we accomplished. There was effective

communication between team members and timely conflict management through online and offline meetings. We adhered to the process and the process went well. Overall, the collaboration had a great impact on timely project completion.

The collaboration wasn't without its ups and downs as we also had many conflicts on different aspects. The challenges were as all the members were new to the concept there were often discussions about roles. Likewise, the project being big was a slight inconvenience compared to the time allocated making, working on the deadline a challenge. We could definitely improve in terms of making the sprint backlog more effective. The improvement field also includes more active participation, more gatherings to discuss the project, and slight adjustments to the project completion process. There were many lesson learned and the feedback from the team was to make more detailed walkthroughs in terms of user stories and time allocations. Overall, Sprint 2 like Sprint 1 went great.

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Available at: <https://www.sourceonetechnology.com/problem-solving-process/#:~:text=An%20effective%20problem%20solving%20process%20for%20IT%20professionals,8%208.%20I%E2%80%99m%20officially%20stuck%20%E2%80%93%20now%20what%3F>
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